



CHNT

Empower the World

Introduction of CHINT LV R&D

November 2025

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TamirZMubarrad 2146

1) Introduction

40+ years of history

Focusing on product
and creating a CHINT
brand with one mind



CHINT

Empower the World

2016~Now

Building a platform
empower innovation

Science and Technology Innovation Park |
Research Institute | One Cloud, Two
Networks

2006~2015

Green energy
intelligent manufacturing

Photovoltaics | Automation | Listing | Intelligent Manufacturing

1984~2005

Adhere to industry
integrate development



Startup | Establishment of Chint Electric Appliances | Transmission and
Distribution

CHINT is committed to driving sustainable development

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Today's CHINT Group

A trusted global leader in innovative smart energy solutions

22 Billions USD

■ ■ operating ■ ■
revenue

140+

■ ■ Countries and regions ■ ■
with business
coverage

22 Years

■ ■ Continuously listed ■ ■
among the top 500
Chinese enterprises

50000+

■ ■ Global ■ ■
employees

500 Thousands

■ ■ Drive employment in ■ ■
the industrial chain

25 Billions USD

■ ■ Total assets ■ ■

Business

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The CHINT group covers key sectors like: smart electrical, green energy, industrial control, automation, smart homes, and smart metering. We have built a complete industry chain that includes power generation, storage, transmission, distribution, delivery, usage, and monitoring

Smart Electrical System



Green Energy



Smart Home



Industrial Control and Automation



Building
electrical appliances



Energy efficiency
management



Intelligent
manufacturing



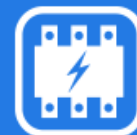
Automation



Smart heating



Smart water
management



Low Voltage
Apparatus



Power transmission
& distribution



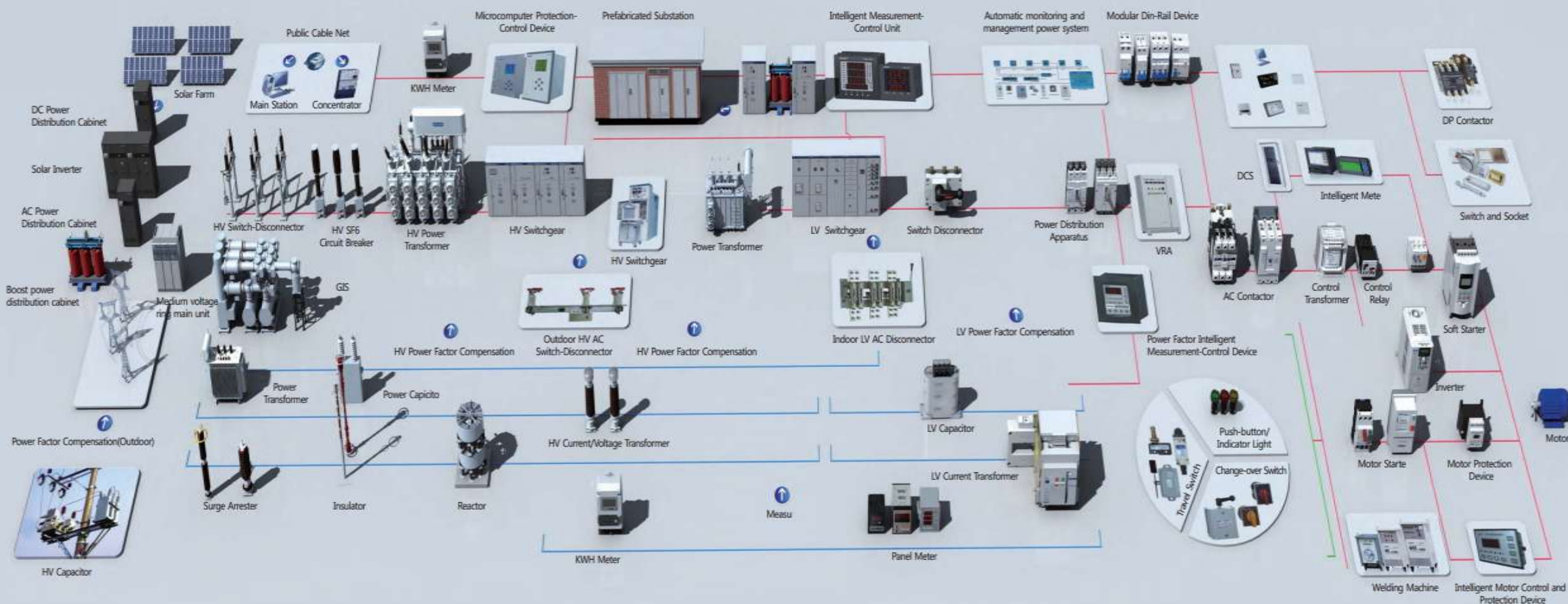
Instrumen-
tation



Smart
Home



Smart
building



Through last over 40 years development, CHINT has been expanding product line in electric industry field. Today CHINT can provide almost all products in entire electrical industry from the Generation, to the Transmission, and to the Distribution.

2) CHINT LV Intelligent Electrical Research Institute

1- WAA R&D Team

2- R&D Capability

CHINT Intelligent LV Electrical Research Institute

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Specialize in technology research and the development of intelligent low-voltage electrical products, like ACB, MCCB, MCB, industrial control switches, drive products, switchgear, smart systems, and related electronics.

- 1) Headquarter is in Shanghai
- 2) Xi 'an has a branch of electronic technology department
- 3) Hangzhou for the IoT technology research branch
- 4) Four overseas R&D centers



At the heart of CHINT's R&D Center is a dedicated team of over **1500** researchers and engineers.

Over the years, our team has successfully developed thousands of products that meet the highest standards of quality and performance.

With more than **10,000 patents**, CHINT leads the way in intellectual property and technological advancement.



Thousands of Products

+1,500 Researchers Engineers

+10,000 Patents

Statistics:

- Over 12% of our R&D engineers hold a PhD or a Master's degree.
- 52% of engineers are at middle and senior levels.
- 15 leading technical talents.
- CHINT invests up to 12% of its annual revenue in research and development (R&D).



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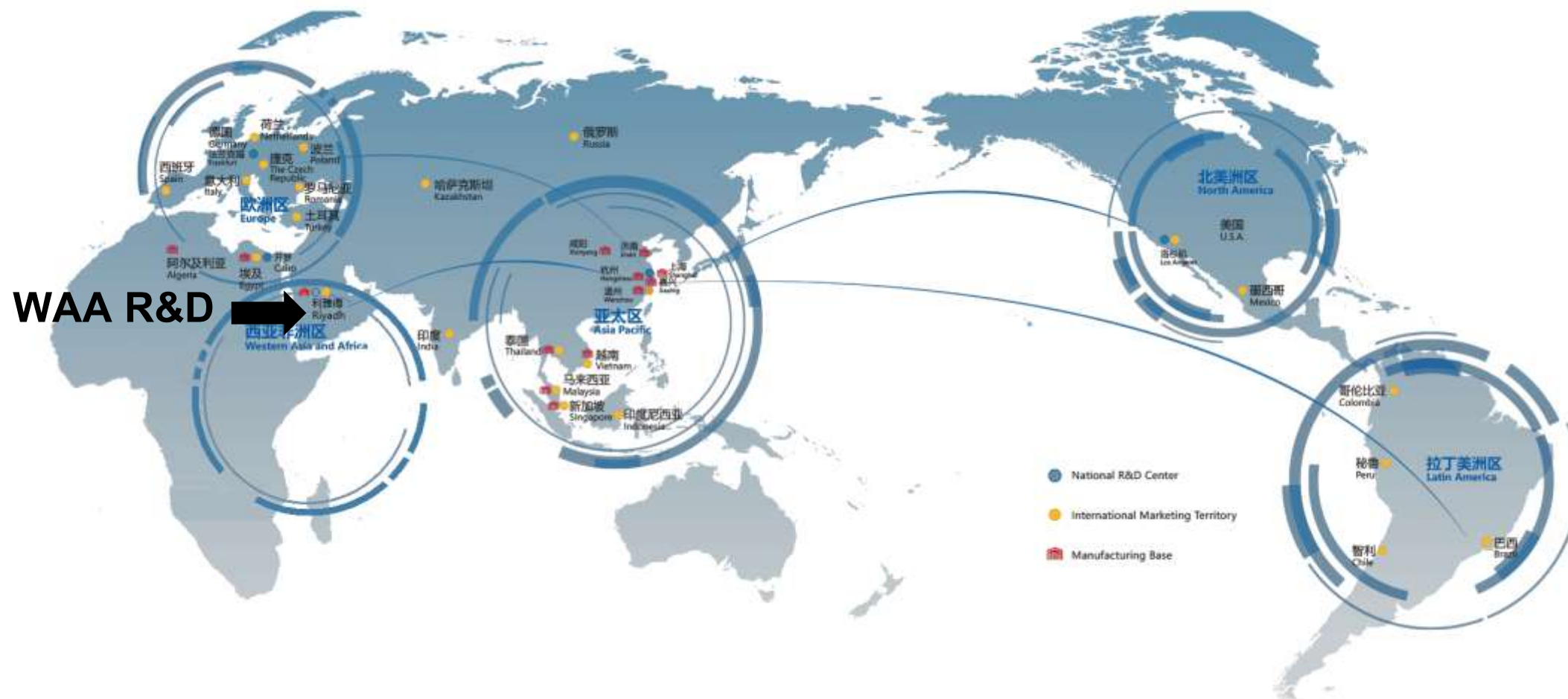
WAA R&D Team

West Asia and Africa (WAA) R&D Department

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- CHINT established this R&D department in Riyadh to serve WAA region, with a focus on Saudi Arabia



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West Asia and Africa (WAA) R&D Department

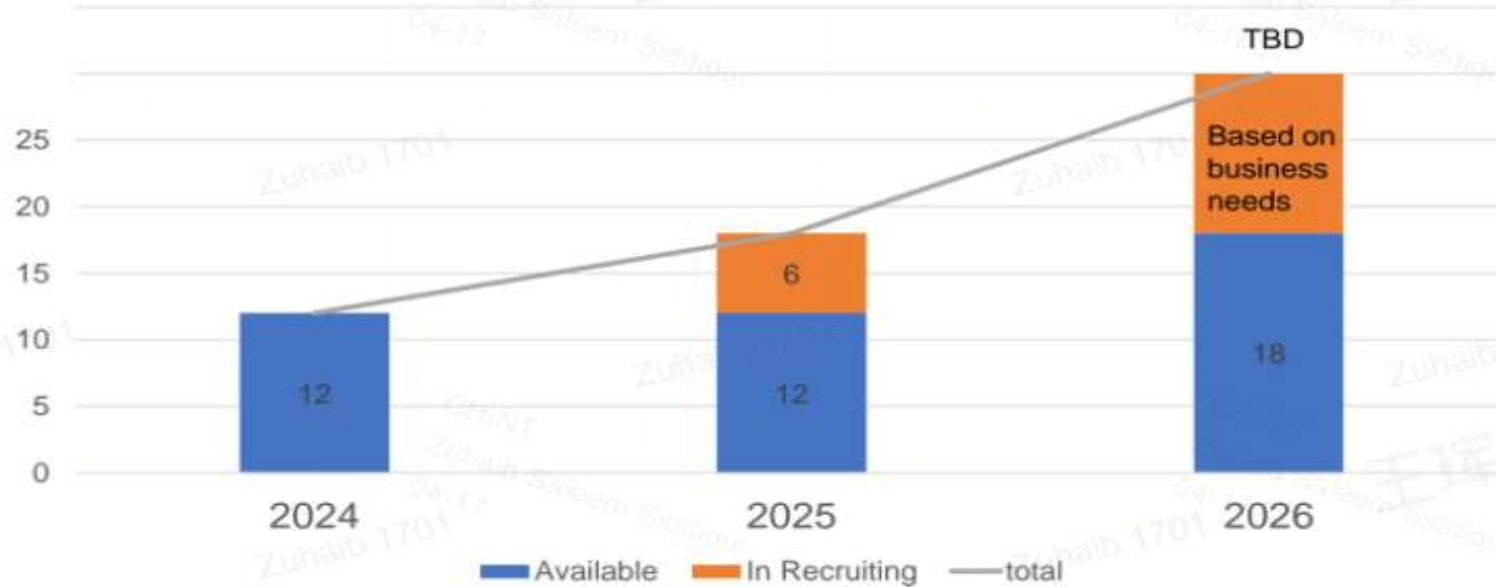
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Job Positions:

No.	WAA DC R&D Position
1	Head of WAA
2	Sr Electronic R&D
3	Sr Energy Management R
4	Components R&D
5	Driver Application R&D
6	LV Switchgear R&D(Syste
7	Certification Engineer
8	Sr Electronic R&D(HW)
9	Sr Electronic R&D(SW)
10	Sr Mechanical R&D
11	LV Switchgear R&D(ME)
12	Sr Electronic R&D(System

WAA R&D Headcount Roadmap



Why a Local R&D Team in Saudi Arabia?

Fast Product Customization:

Efficiently adapt products to meet specific market demands with localized, flexible delivery.

Quick Understanding of Local Needs:

Rapidly grasp customer requirements and provide R&D support, including solution design and demos.

On-Site Support:

Provide fast problem-solving and direct support to ensure smooth project execution.

Building Strong Local Partnerships:

Maintain strong ties with key market players, like our success stories with SEC (Saudi Electricity Company).

R&D Capability

输出功率: $P_{OUT} = U_{OUT,1} \cdot I_{OUT,1} + U_{OUT,2} \cdot I_{OUT,2} = 12 \text{ W}$

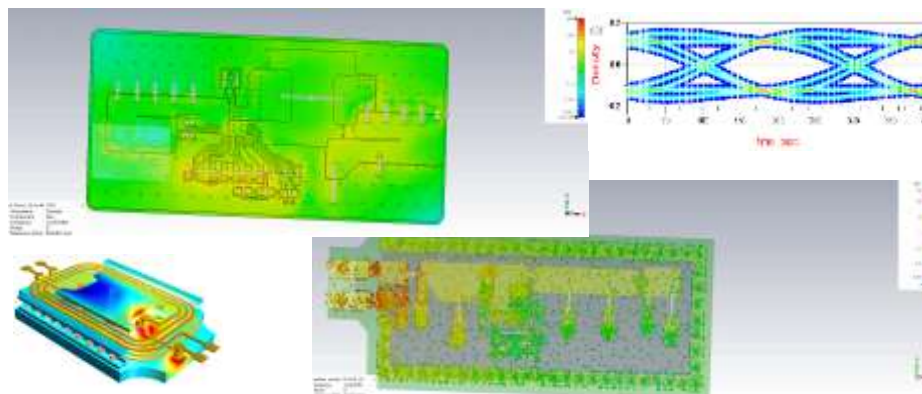
输入功率: $P_{IN} = \frac{P_{OUT}}{\eta} = 14.12 \text{ W}$

输入线电压峰值: $U_{Hmin} = U_{Nomin} \cdot \sqrt{2} = 141.42 \text{ V}$

单工频电容放电时间: $A1 = \frac{U_{ICmin}}{U_{Hmin}} = 0.42$

工频电网爬波时间1: $\sin\left(2 \frac{\pi}{T} \cdot t_1\right) = A1 \xrightarrow{\text{solve}} t_1 \begin{bmatrix} 8.6053283194321453047 \cdot \text{ms} \\ 1.3946716805678546953 \cdot \text{ms} \end{bmatrix}$

给t1赋值: $t_1 = 1.66 \text{ ms}$

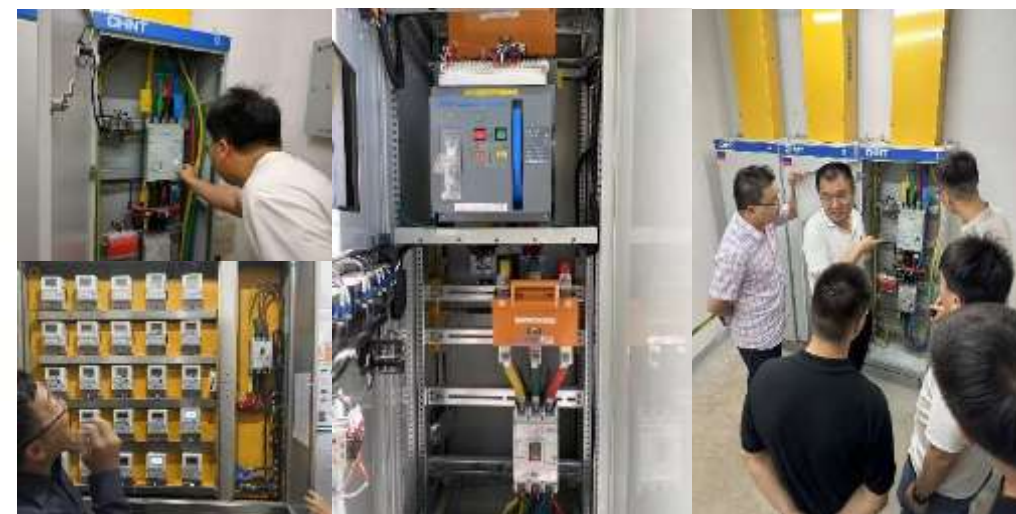


Scientific Foundation and Design Calculations: A modular design approach for power supplies, EMC, and signals, where critical components and circuits are carefully chosen to ensure durability, long product life, and clear traceability of the design.

Simulation Analysis: Utilize EDA tools (CST, HYPERLYNX, LT Spice, PSIM) to evaluate and compare design options, choose the optimal solution, and store designs in our shared library for easy access and collaboration.

Precision Testing: Conduct white box testing to assess signal integrity and black box testing to evaluate stability and reliability, ensuring the creation of durable and reliable hardware.

Customer Delivery: Address market needs, build on lab PASS results, and continuously improve for reliable performance in complex environments.

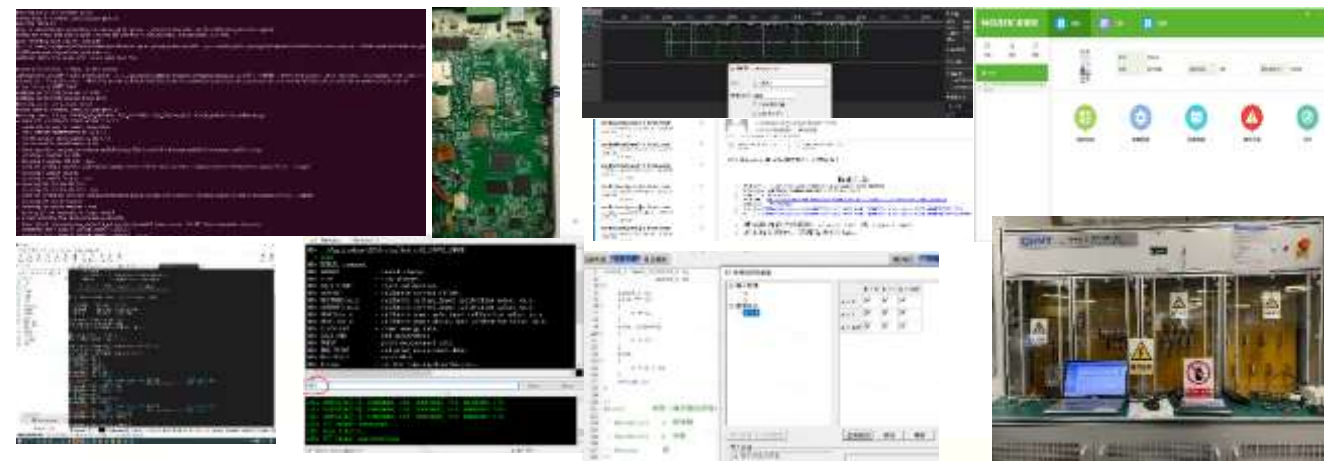


Embedded Software Design Capabilities

- Multi-platform and cross-platform development capabilities
- Software layering, defensive programming, and modular design capabilities
- Online and remote debugging capabilities in multiple ways
- Static analysis, unit testing, integration testing, system testing, HIL and other reliability assurance capabilities
- Equipment modeling, reusable code library, universal software unified architecture for the power industry, automatic code generation, communication enterprise standards and tools, etc. Ensure agile development capabilities



Software Design: Quantitative, Timing, Qualitative
(Architecture - System - Task Design - Space Design
- Timing Design) --- Design Determines Quality



Debugging: GCC, IAR, Keil and other multi-platform, cross-platform development and debugging;
Testing: automatic unit static testing and analysis, and dynamic error finding; system integration testing and product quality verification

System Application Software Design Capabilities

- Independent research and development of equipment cloud platform
- Equipment description file standardization, massive equipment access which can be customized according to user needs
- Platform design allows for rapid integration under high cyber security



System design and review: clarify goals, verify reasonability, feasibility, improve quality, reduce risks --- ensure the project is trackable

Software testing: test tooling, script automation (unit testing, integration testing, performance testing, security testing, etc.) --- ensure the delivery of high quality



Norms And Standards: Unified, Quantifiable, Understandable, And Executable—win In Unison

Operation and maintenance monitoring: automation tools, real-time monitoring, accurate positioning --- ensuring system performance stable and reliable

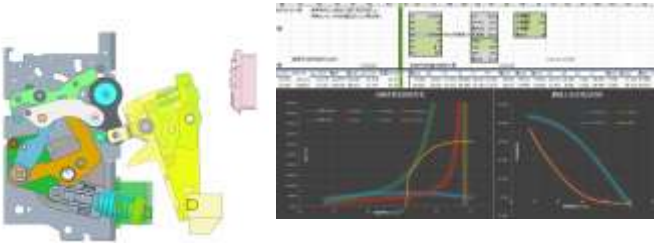
- Simulation platform with **mechanism motion simulation, structural strength simulation, electromagnetic simulation, arc simulation, heat flow simulation, tolerance analysis, power system simulation and more physics coupling** simulation capabilities

Mechanism Motion Simulation

Simulation function: analysis of the mechanism's force/torque, speed, acceleration, opening and Closing time and other dynamic characteristics, mechanism interference inspection; theoretical

Calculation and optimization design of mechanism size, dead zone and force.

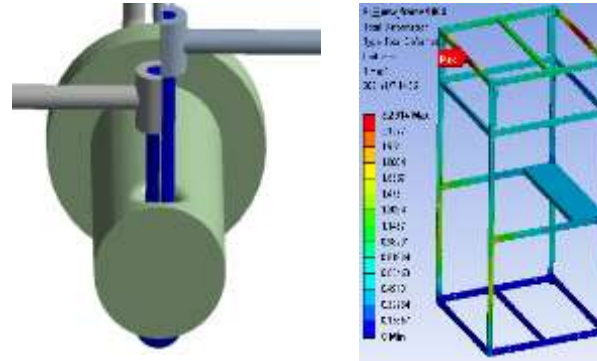
Simulation object: operating mechanism and other motion mechanisms



Structural Strength Simulation

Simulation function: analyze the stress, strain, deformation, contact reaction force, etc. Of parts, and optimize the strength and life of parts.

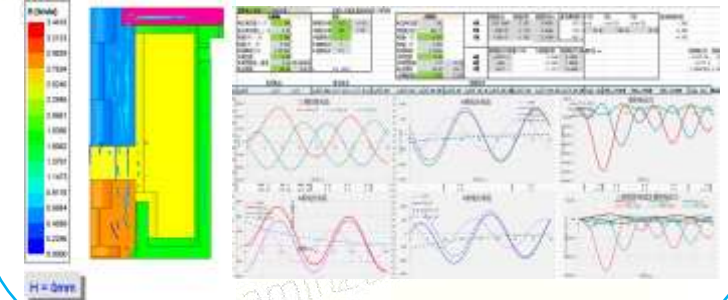
Simulation objects: various plastics, metal structural parts, various springs, etc.



Electro-Magnetic Simulation

Simulation function: analyze the electromagnetic force, electrodynamic force, power consumption, magnetic field distribution, current distribution, and transformer output signal of each magnetic system; magnetic system design, optimization and improvement, coil design and optimization.

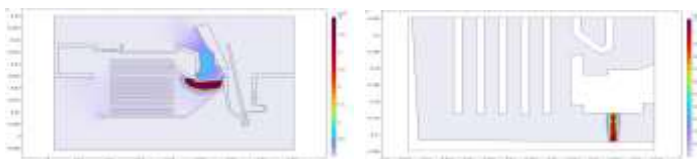
Simulation objects: magnetic releases, electromagnets, contactors, transformers, relays, emr, permanent magnets and other magnetic components.



Arc Simulation

Simulation function: analyze the characteristics of the arc in the contact system and arc extinguishing system, optimize the structural design of the contact system and arc extinguishing system.

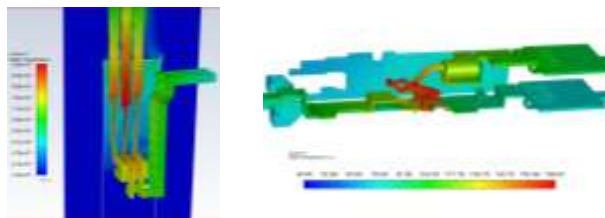
Simulation object: contact system, arc extinguishing system.



Heat Flow Simulation

Simulation function: analyze the characteristics of the arc in the contact system and arc extinguishing system, optimize the structural design of the contact system and arc extinguishing system.

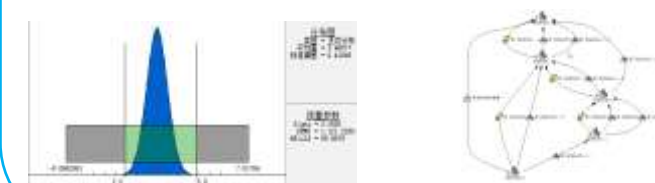
Simulation object: contact system, arc extinguishing system.



Tolerance Analysis

Simulation function: analyze the pass rate of functional dimensions such as buckle amount and overtravel, and optimize the size and tolerance distribution.

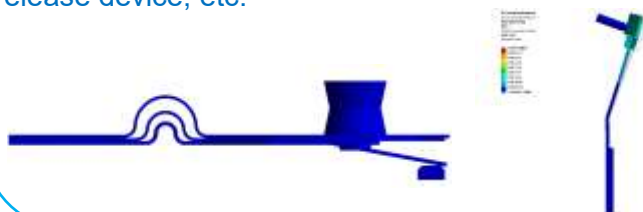
Simulation object: mechanisms with special requirements for tolerance fit



Multiphysics Coupled Simulation

Simulation function: analyze the coupling problems between different physical fields such as electromagnetic - structure, electromagnetic - mechanism, electromagnetic - thermal - structure, and more accurate analysis of engineering problems.

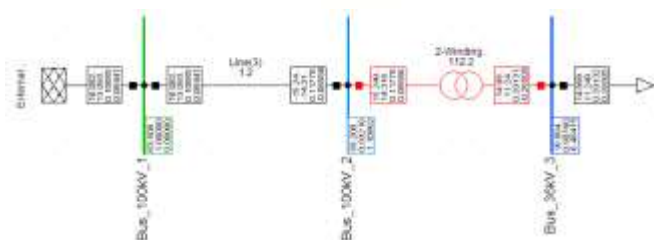
Simulation objects: thermal release device, magnetic release device, etc.



Power System Simulation

Simulation function: analyze the characteristics of new power system power flow and short-circuit current at different short-circuit points.

Simulation object: power system



Power System Simulation

High-performance computing cluster:

5 nodes RAM: 768G CPU: 128-core

10+ Xeon scalable dual-core workstations.



Process Engineering Center

It covers all the processing capabilities of low-voltage electrical appliances, and can adopt the optimal process plan according to the requirements of different R&D stages, through mechanical verification, reliability verification, customer testing, customer technical exchange and other different prototype processing.

Multiple Sets Of Special Equipment

Existing CNC machining center, 3D printer, engraving machine, Laser cutting machines, lathe and milling machines, spring machines and other special equipment.



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CHINT Intelligent Electrical Laboratory Center

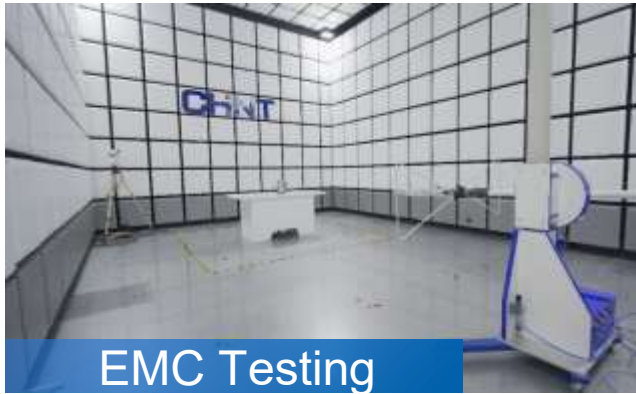
A comprehensive laboratory platform integrating R&D process experiments, multi-standard product type tests, and field application simulation experiments, with a total construction area of 9,600 square meters. With it, we'll be able to offer high-quality R&D, prompt project execution and meet end-to-end custom-tailored delivery for global customers.

Three Power Supply Systems

With 350MVA@50Hz/60Hz short-circuit generator set, 1600kVA@60Hz generator set, 35kV@50Hz special line for power grid, the laboratory center can meet the requirements of all-weather short-circuit and , short time withstand current test.



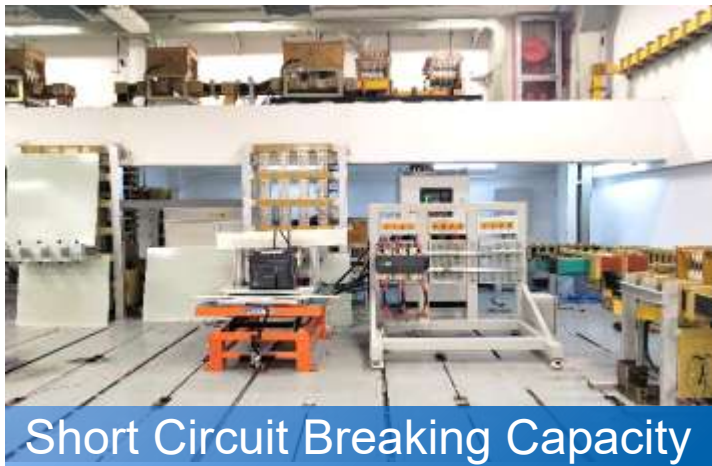
Extensive Test Labs



EMC Testing



3D X-ray Testing



Short Circuit Breaking Capacity



Industry Application Simulation

Covers the **GB/IEC/UL standard** system for low-voltage electrical appliances and can meet the program experiments of national standards, international electricians, and American standards. Core experimental capabilities such as short-circuit breaking capacity test (150kA@480V), radio frequency radiation test(3m-Anechoic Chamber), 3D X-Ray, and Field Application Simulation Experiment scenarios can **meet customers' requirement of multi-dimensional technical experiment.**

Short-circuit Test

AC 50Hz
150kA

AC 60Hz
100kA

DC
100kA

Elec. Endurance Test

AC 50Hz
10kA

AC 60Hz
6.3kA

DC
10kA

Icw Test

AC 50Hz
480V/150kA/1s,
100kA/3s

AC 60Hz
508V/100kA/1s

DC
1000V/100kA/1s

EMC Test

Anechoic Chamber

3m Anechoic Chamber

RF Radiated Immunity

Test Frequency: 80MHz~6GHz,
Electric Field Strength: 54V/m

RF Conducted Immunity

Test Frequency: 150k~230MHz, 18V

Env. Test

- Damp heat
- Thermal shock test
- Rapid temperature change test
- Salt Mist test
- Vibration test
- Enclosure protection test

Temperature-rise Test

AC 50Hz
Three-phase 10kA

AC 60Hz
Three-phase 7.5kA

DC
DC 5kA

Elec. Characteristic Test

Long Time Delay 10kA,
Short Time
Delay/Instantaneous 75kA

AC 60Hz
Long Time Delay 7.5kA,
Short Time
Delay/Instantaneous 50kA

DC
Long Time Delay 5kA,
Short Time
Delay/Instantaneous 20kA

Driving Product Performance Test

• Load Test • Lifting Test • Towing Test • Protection Test

Materials Analysis

• 3D-X-ray • Optical Microscope • Cold Field Emission
Scanning Electron
Microscope

Field Test

Primary and secondary
fusion test, customized test
for customer needs

Empowered With Smart Interoperability

We realize experimental witness, remote audit, and technical discussion for customers through centralized monitoring and control of the experimental system, visual presentation of process data, combined with remote interactive sharing.

LIMS



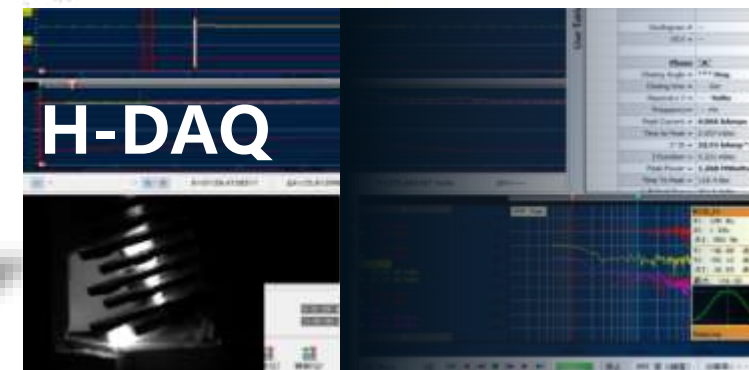
ALM



LDDBS



H-DAQ

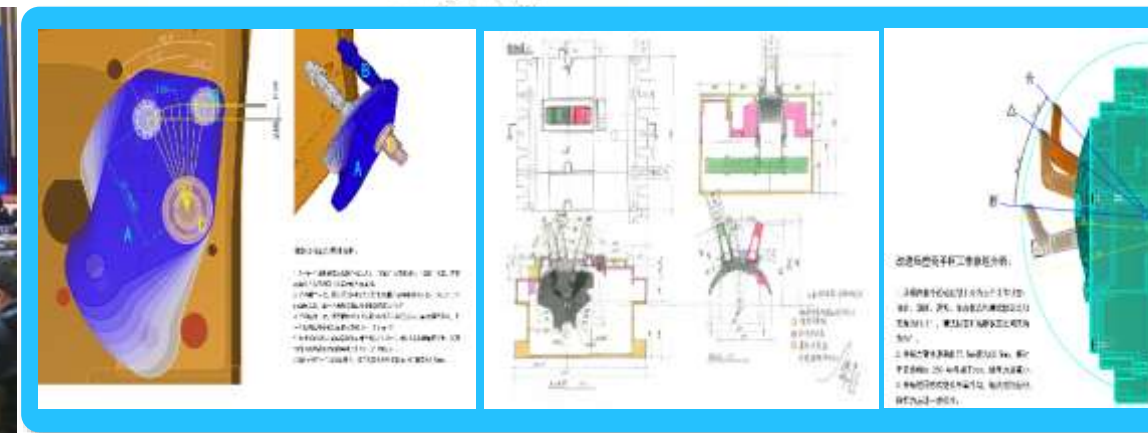


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- We have a professional team to provide all-round design capabilities such as shape design, UI design, video, animation, digital twin, multidimensional modeling, packaging design, etc.
- Strong internal skills, emphasis on cooperation, the use of advanced design theories, the formation of "human-machine interaction", "color environment" and "visual effects" a series of research results, consolidated the theoretical foundation of Chint industrial design



R&D Capability_Industrial Design 2

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- Theory + practice, through structural optimization, image unification and other series of work, in the modeling, color matching, logo, UI, interaction, 3D animation, video editing and other aspects, the formation of CHINT's industrial design ecology, typical products and image identification solutions



	正泰标准系列	正泰昆仑系列	诺雅克标准系列	诺雅克其他系列	电力专供系列	住宅专供系列
系列	8系列、1系列、老产品	NX系列	Ex9系列	北美产品等	NBP	NBH
造型						
配色						
标识						

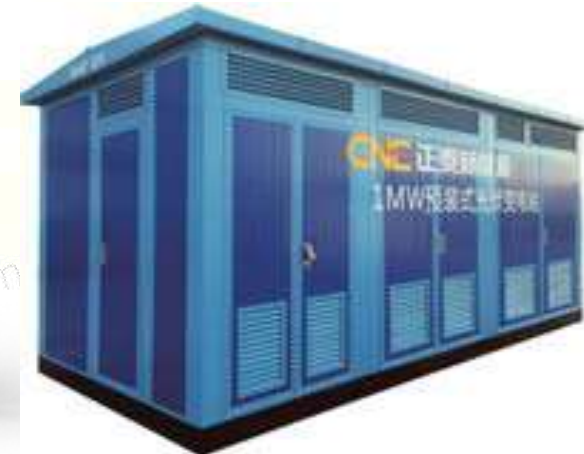
3) Projects and Products Supplied to SEC

3) Products Supplied to SEC (Saudi Electrical Company)

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CHINT supplied to SEC:
GIS, RMU, ECB, MCCB, ACB,
Switchgears,
Transformers, & other customized
products made specially for them.



pictures of some products
that were supplied to SEC.

NG7 17.5 GIS_Background

CHINT

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NG7 is a new SF6 gas insulated switchgear independently developed and designed by CHINT.

- It can be used in primary and secondary power system;
- Rated voltage from 10kV to 35kV,
- Rated current from 630 to 3150A.
- The primary high voltage components be completely sealed in the gastight welded stainless-steel tank, and then filled with SF6 gas as insulation medium which has good insulation and arc extinguishing performance.
- All solid insulation plug cable termination connection, safe and convenient.



**NG7-12~40.5kV
Gas insulated switchgear**

Product features :

- 🏠 Lifetime maintenance-free design;
- 🔧 Gas insulated, compact structure;
- 🌍 Can withstand adverse environmental effects;
- 📦 Convenient expansion;
- 🔵 Safe and reliable performance;
- 🔋 Economical and environmentally friendly



NG7-17.5 GIS_Technical Bright Spot

Main parameters full fill

Rated Voltage : 17.5kV;
Rated Current : 200\400\630A;
Rated short-circuit breaking current : 21kA ;
Rated short-time withstand current : 21kA/4s ;
Altitude : 2000m ;
Protection grade: Gas Tank: IP67. Enclosure: IP54;
Overall dimensions :
Width * depth * height (mm) 1380/975/1850 (2L1T)

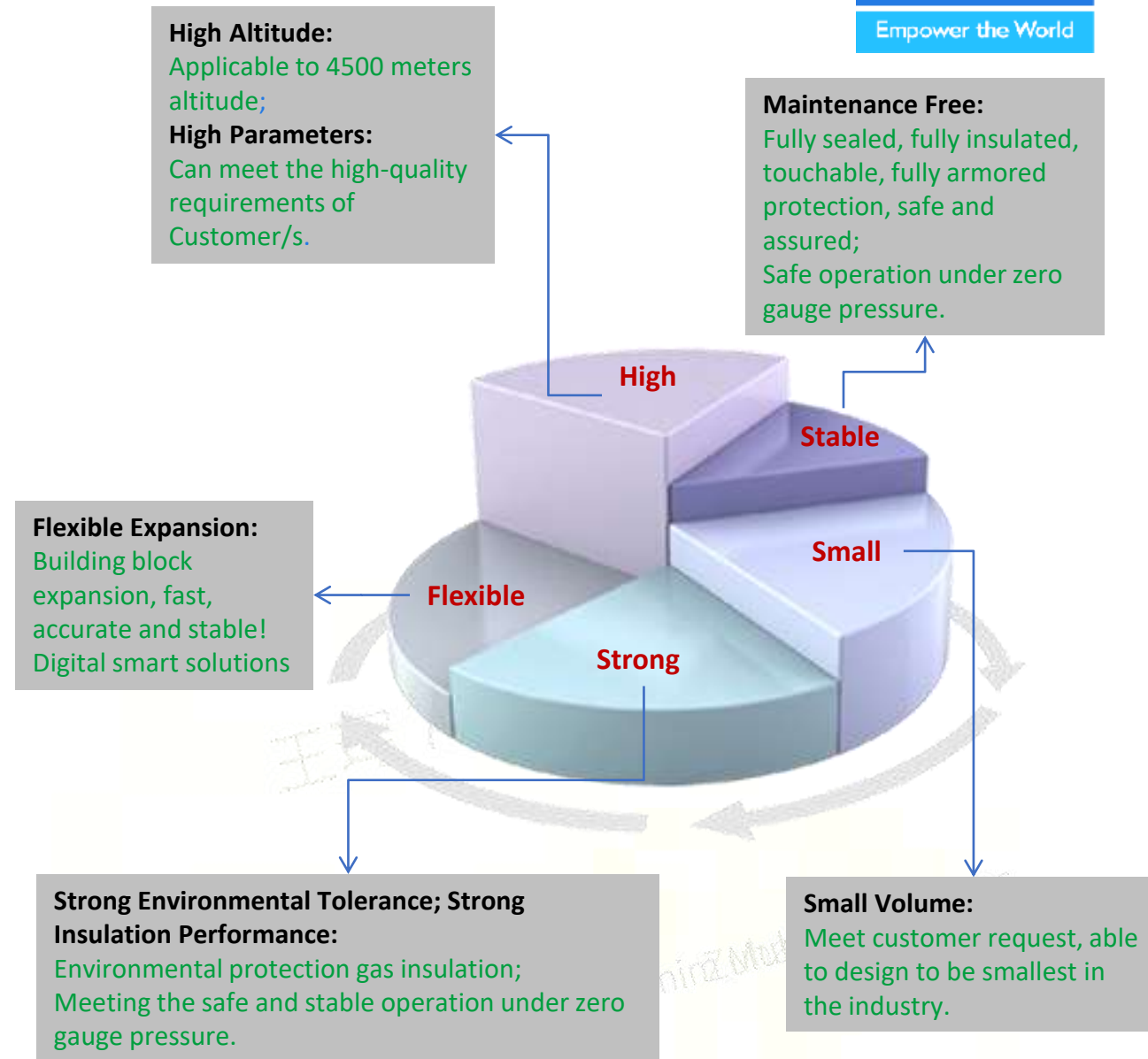


NG7-17.5kV 21kA

Gas fully insulated metal enclosure switchgear

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Oil-immersed Power Transformer

No.	Country	End User	Product Type	Qty.	Ordering Time	Delivery Time	Project Name
1	Saudi Arabia	Saline Water Conversion Corporation	S-2500/4.16	4	2015-09-28	2015-12-28	Hyflux Red Sea project
2	Saudi Arabia	Saline Water Conversion Corporation	SZ-11000/13.8	4	2015-09-28	2015-12-28	Hyflux Red Sea project
3	Saudi Arabia	Saline Water Conversion Corporation	ZS-630/4.16	2	2015-09-28	2015-12-28	Hyflux Red Sea project
4	Saudi Arabia	Saline Water Conversion Corporation	S-3200/4.16	2	2015-09-28	2015-12-28	Hyflux Red Sea project
5	Saudi Arabia	Saline Water Conversion Corporation	SCB10-600/11	2	2015-09-28	2015-12-28	Hyflux Red Sea project
6	Saudi Arabia	Arab Steel Co.	SF11-45000/22	1	2011-10-17	2012-02-17	Saudi Arab Steel factory project

LV Switchgear Panel NGL2

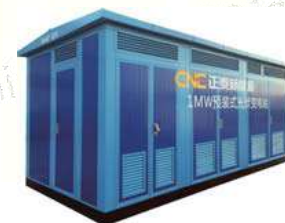
1	Saudi Arabia	Saline Water Conversion Corporation	NGL2	6	2015-09-28	2015-12-28	Hyflux Red Sea project
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Prefabricated Substation YBM(P)29

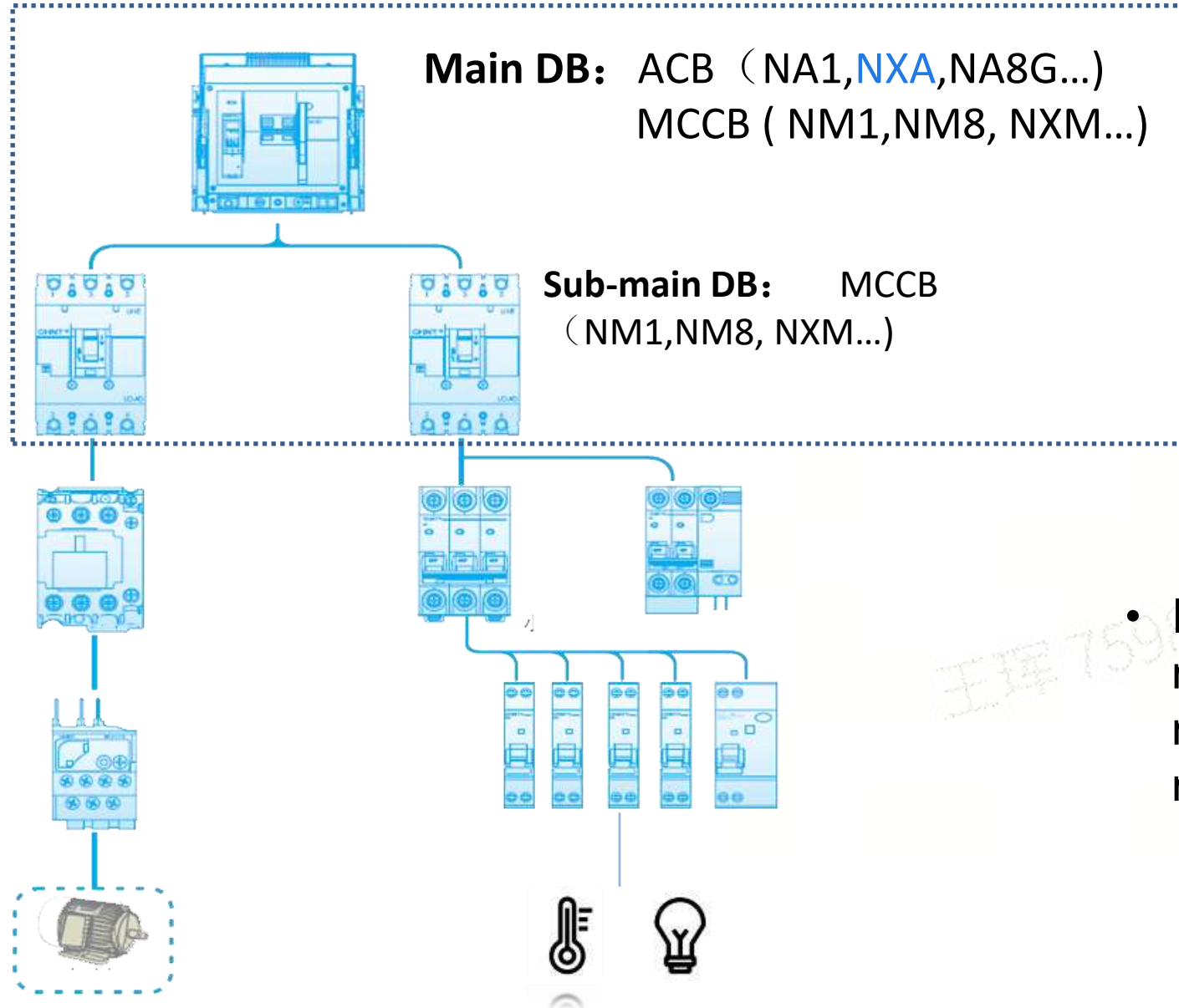
1	Saudi Arabia	Saline Water Conversion Corporation	YB29-0.4	12	2015-09-28	2015-12-28	Hyflux Red Sea project
2	Saudi Arabia	Saline Water Conversion Corporation	YB29-12	5	2015-09-28	2015-12-28	Hyflux Red Sea project

DC Panel GZDW

1	Saudi Arabia	Saline Water Conversion Corporation	GZDW-300AH-110V-G	2	2015-09-28	2015-12-28	Hyflux Red Sea project
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Tami 2146



- In 2020, CHINT got request from SEC for ACB requirement, NXA was recommended

KEMA
EUR**CE**

NXA Series

- In from 400 to 6300A
- Max. I_{cu} up to 120kA
- I_{cw} up to 100kA/1s
- 3P 4P available
- Use 415V/690V for AC 50/60Hz
- Draw-out type / fixed type
- Manual or electrical
- Standard: IEC/EN 60947-2

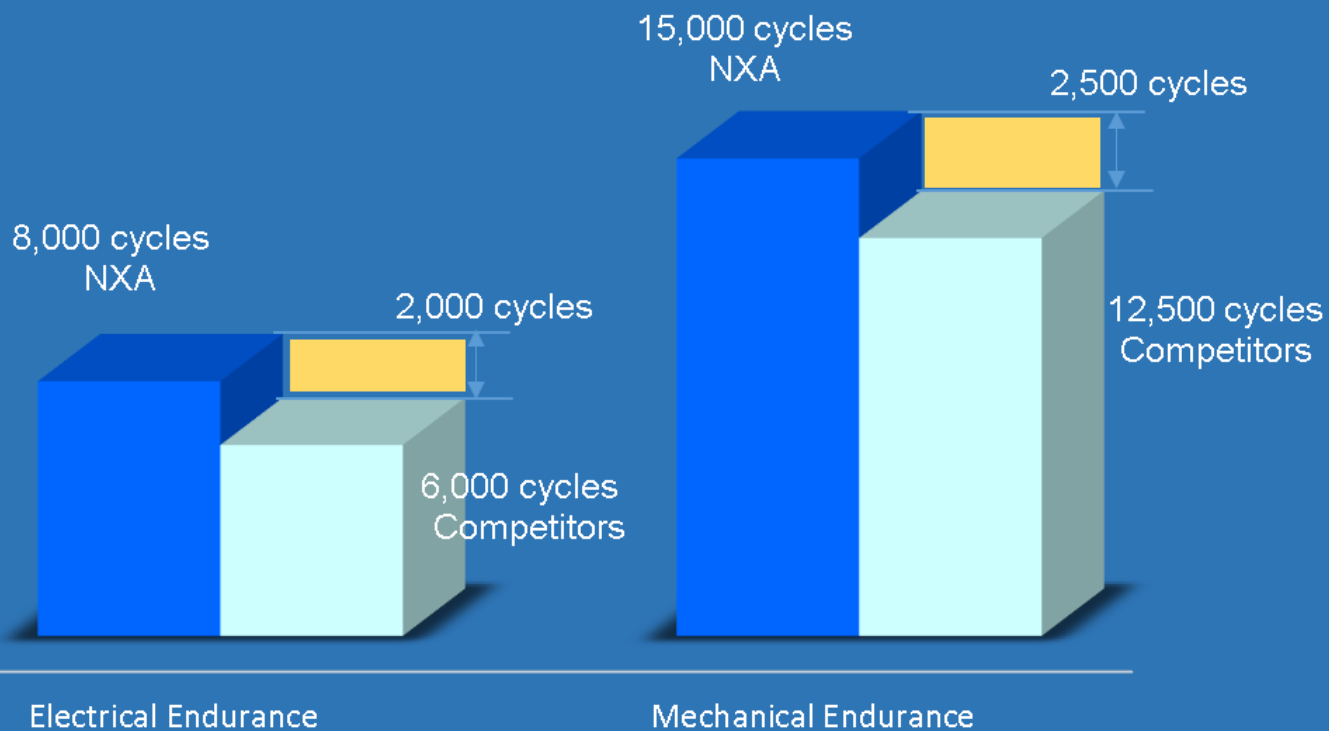
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Longer

High Endurance

Elect. 8,000 cycles
Mech. 15,000 cycles



Safety and
reliability



Environment
adaptability



Complete
accessories



Digital
experience



In actual application, NXA is Reliable and adaptable to high ambient temperature, +55°C.

- NXMG customized developed for Saudi Market
 - equipped with cage terminal blocks, easier for users to connect.
 - test tripping button added on the basis of locking system, greatly improve product safety
- Standards: IEC 60947-2, 37-SDMS-01, 37-SDMS-02, 37-SDMS-03
- Setting temperature: **+55°C**



NXMG - 125 C □

Rated current

Blank: No lock;
C: With lock

Frame size rated current:
125A, 250A, 400A, 600A

Improved type (Apply to Saudi market)

Product model :
NXM series moulded case circuit breaker

- Overall power consumption of the product is reduced, and the reliability is greatly improved.
- Freely connected up and down, and the wiring method is more flexible.
- Anti-electricity theft design
- Key lock function design
- Innovative use of bearing structure
- Reshape the mechanism to eliminate slipper problems and improve product stability;
- Innovate the current limiting structure to improve the breaking reliability of small-sized products;
- Standard protective cover to match the high temperature and dusty environment of the Saudi market;



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MCCB



NM1 series



NXM series



NM8N series



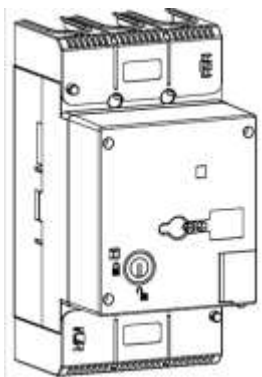
NM1, NXM -

The power can only come from the top.

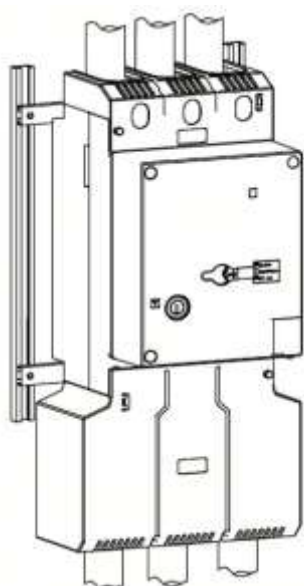
NM8N -

The power can both from the top and from the bottom.

- CHINT got request from SEC for MCCB requirement, NXMD was customized developed.
- Application field: Installed inside meter box, with smart meter controlled to fulfill Payment Control purpose



125-250A



300-400A

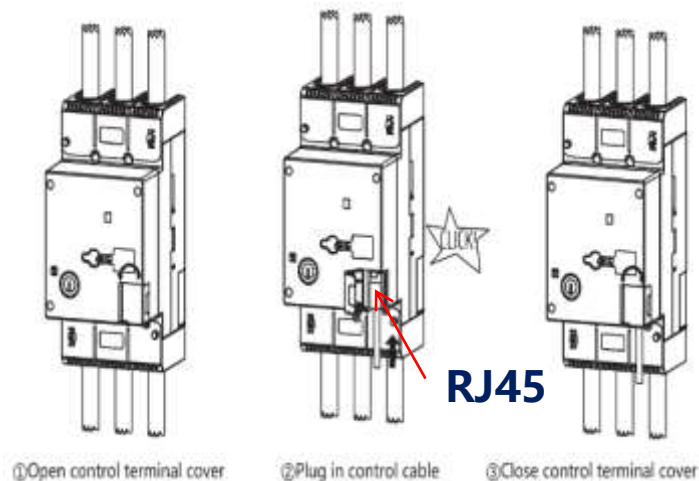
- Range from 125A to 1000A, 4 frames
- remotely switch on/off, compacted size in depth
- Self-powered without external power supply
- control and monitor breaker status can be used Modbus communication
- Rated Temperature at 55°C
- Thermal-magnetic protection
- Short-circuit protection

Frame size Inm (A)		250	400	600	1000
Rated current In (A), 55°C		160、200、250	300、400	500、600	800、1000
Rated insulation voltage Ui (V)		1000	1000	1000	1000
Rated impulse withstand voltage Uimp (kV)		8	8	8	8
Rated operational voltage Ue (V), AC 60Hz		230/400	230/400	230/400	230/400
Number of poles		3P	3P	3P	3P
Utilization category		A	A	A	A
Rated ultimate short-circuit breaking capacity Icu (kA)	AC230V	25	36	70	70
	AC400V	20	20	40	40
Rated service short-circuit breaking capacity Ics (kA)	AC230V	25	36	70	70
	AC400V	20	20	40	40
Mechanical life (times)		/	7000	4000	4000
Electrical life (times)		AC400V,In	1000	1000	1000
Arcing distance (mm)		≤50	≤100	≤100	≤100
Shield		■	■	□	□
Locking function		■	■	□	□
Case wiring terminal		■	■	□	□
Lead seal		■	■	□	□

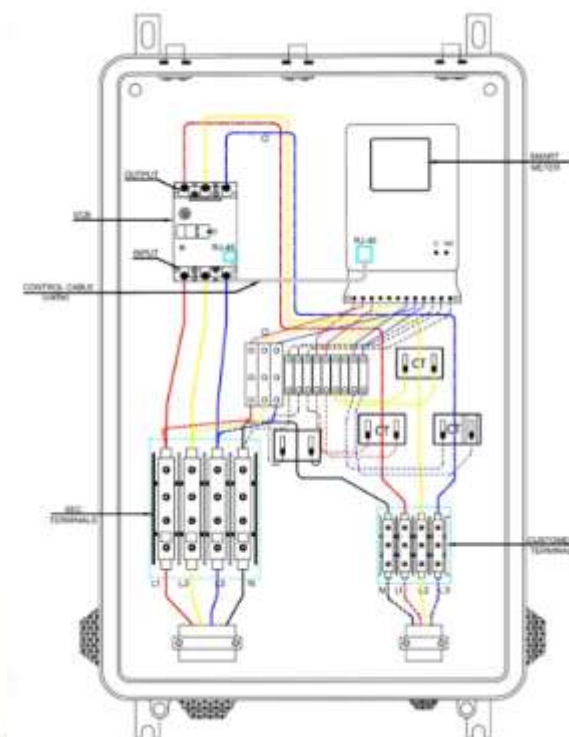


- **Fault Indicator**

Clear status indication



RJ45



Smart Meter send "OpenECB" Command to ECB

Send: 01 05 01 01 XX XX

01-ECB

05-MeterAction

01-Length Of Data

ECBStatus: length 1

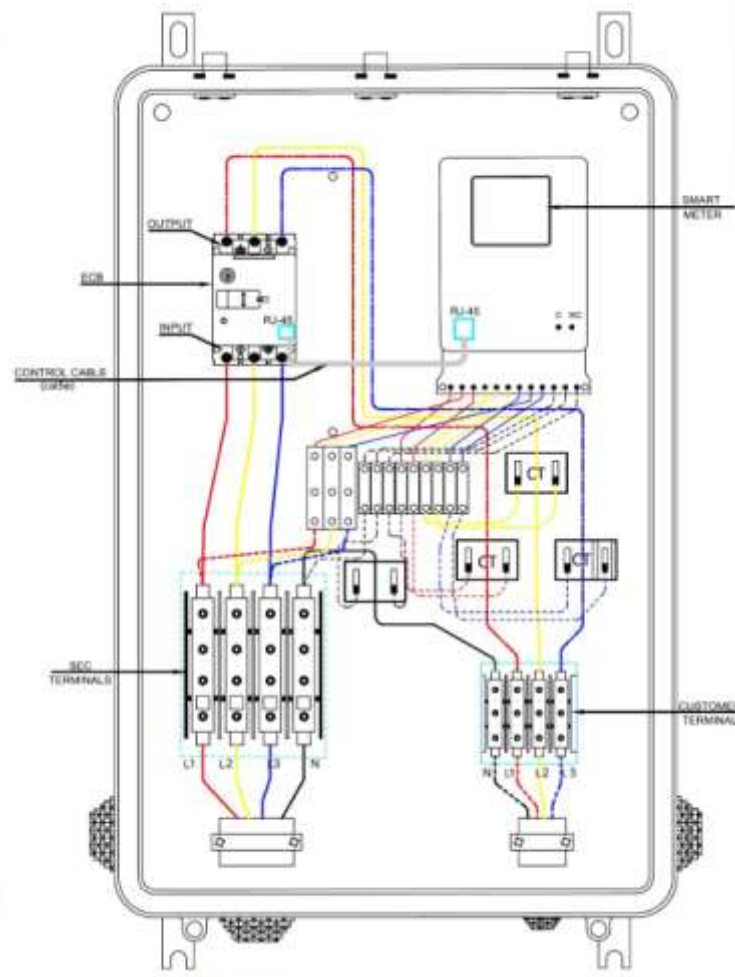
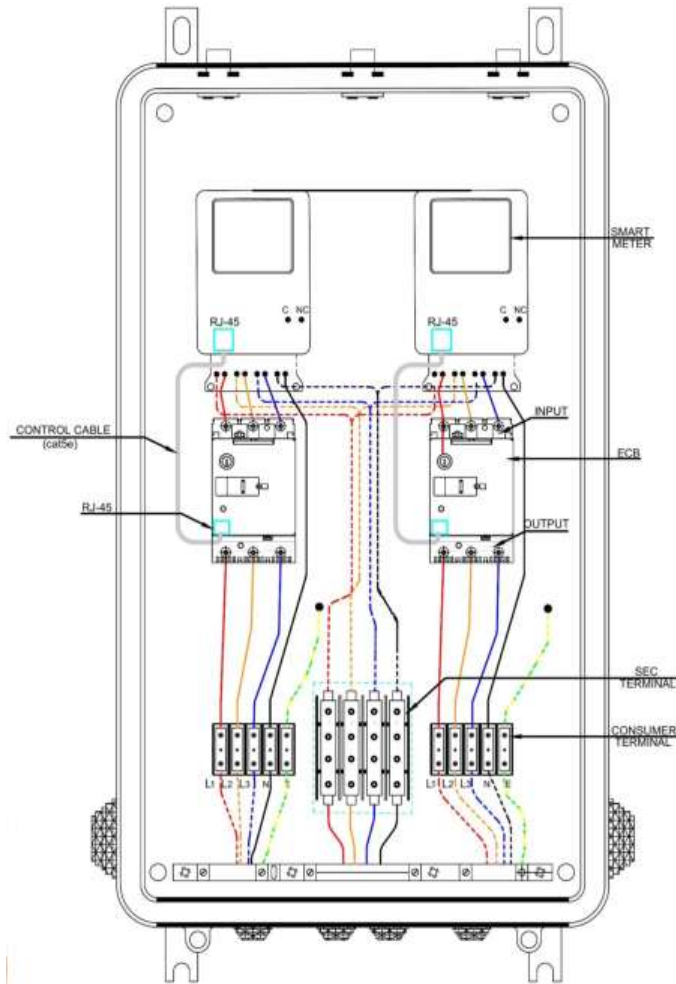
Open: 0x01

Close: 0x02

Tripped: 0x03

Tamper: 0x04

Command	Feedback
3. Client can get ECB status quickly through Modbus communicate between Smart Meter and ECB.	



- Two wiring applications between NXMDs and Smart Meters in Enclosure

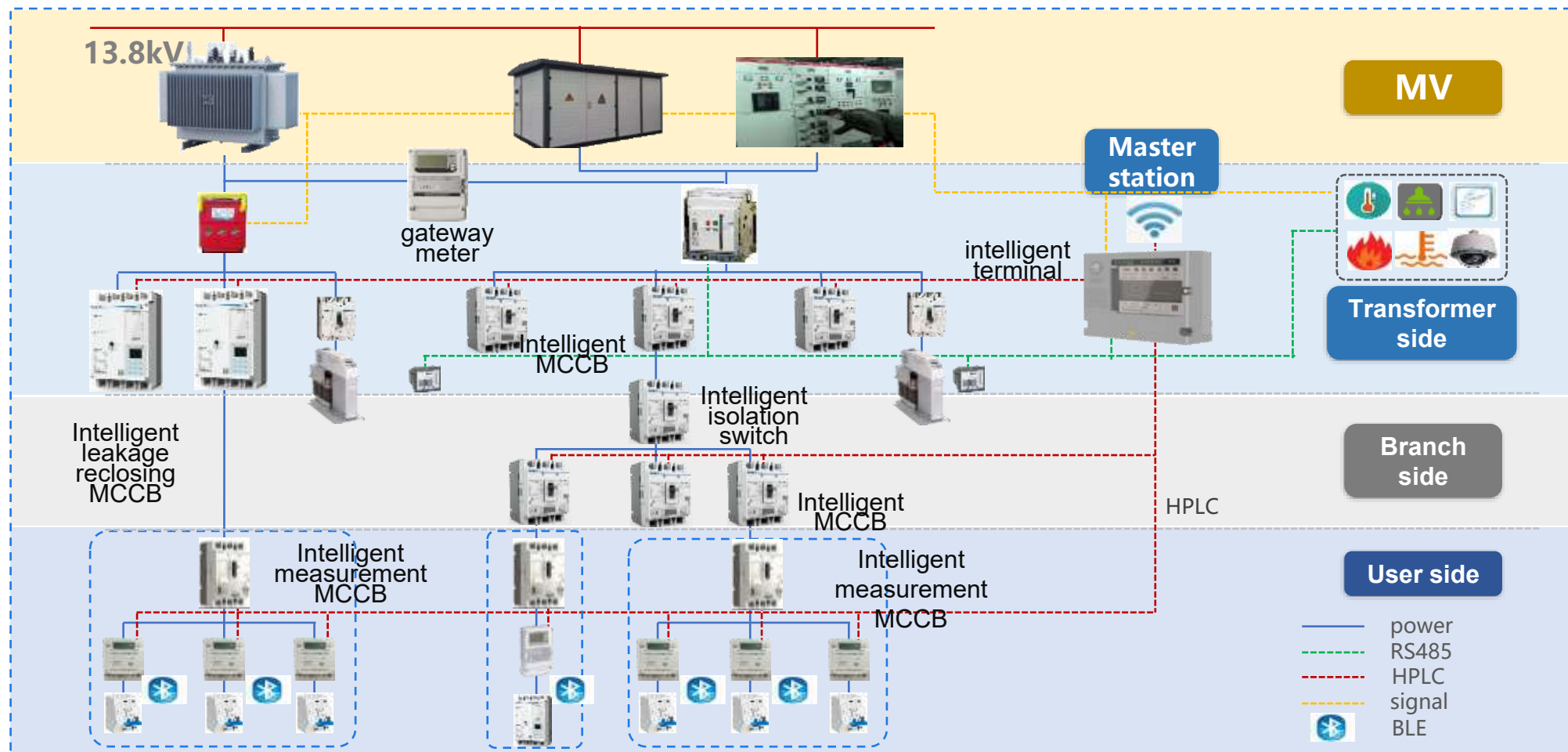
Tamir Mubashir Mad 2146

4) Introduction of New Products and Technologies

Intelligent substation application scheme system architecture

New technology application: precise measurement, topology identification, **BLE** and **HPLC**.

Working with main system, it can provide support for line loss, load management, fault analysis ...

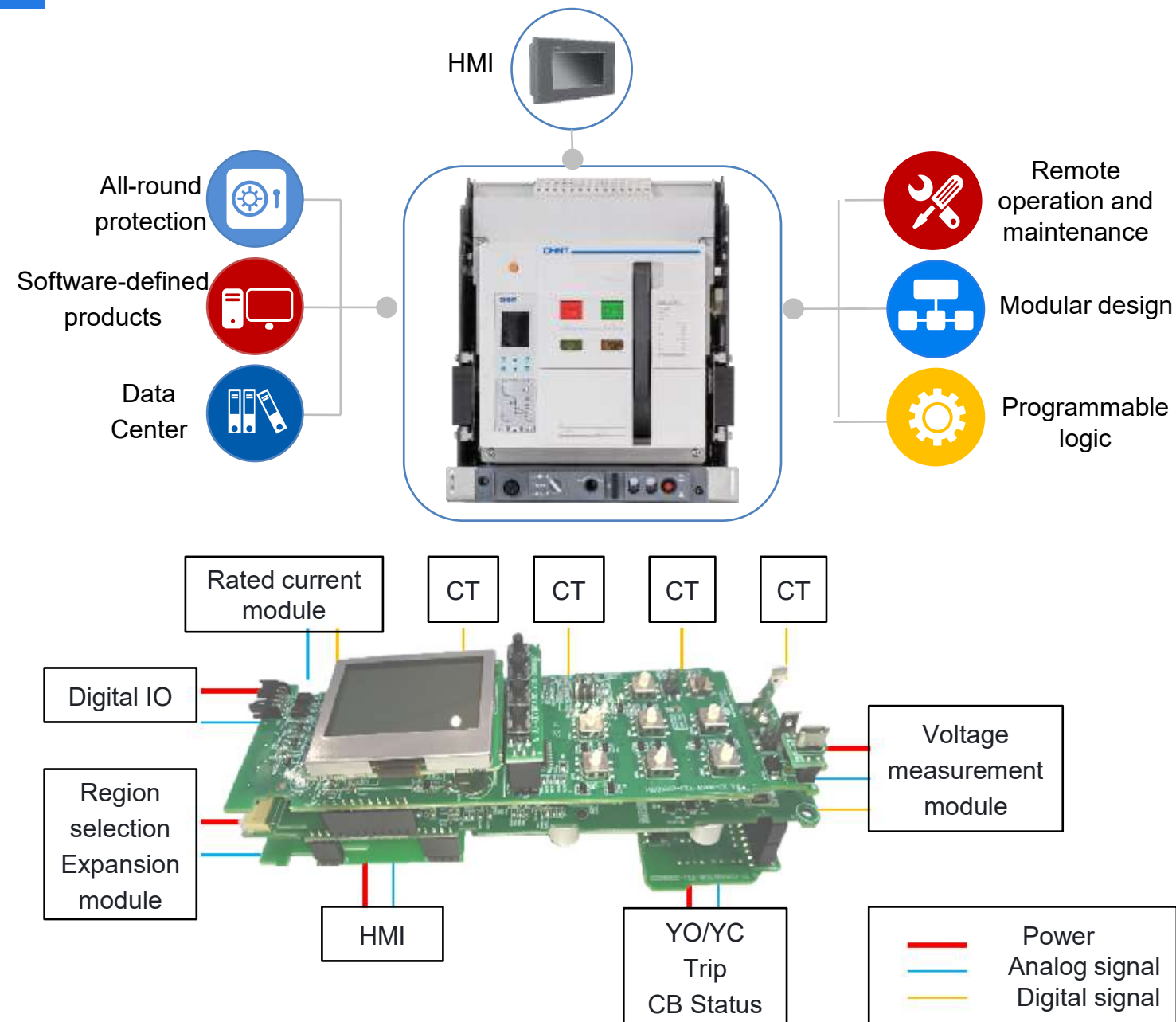


Intelligent substation application scheme

Smart ACB_Features

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• Product parameters

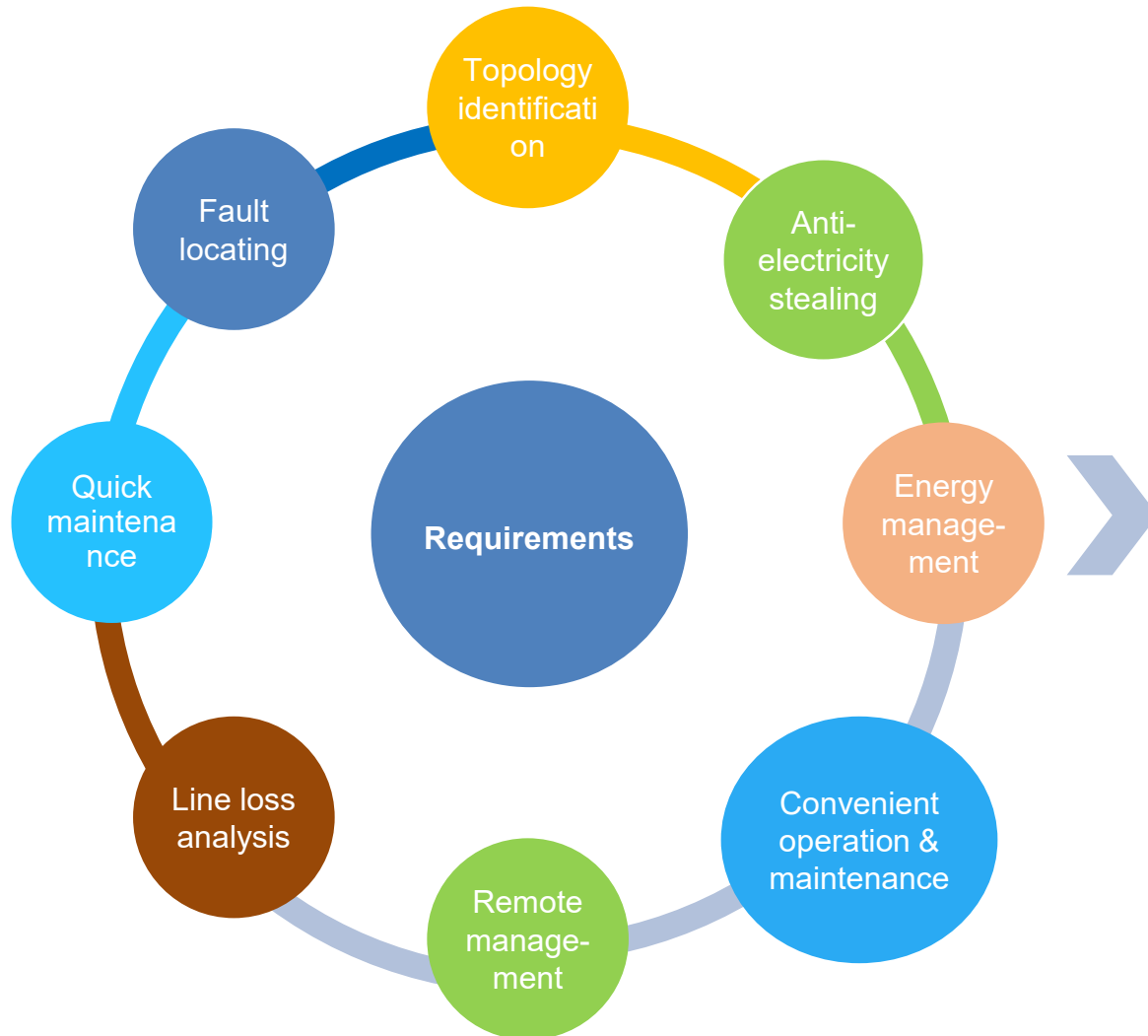
- Poles: 3P, 4P;
- Rated short-circuit breaking capacity: 85 kA ~ 135 kA (55 kA for 1600 A Frame)

• Technical highlights

- The bus can be rotated (vertical or horizontal), wiring is flexible and convenient;
- Control panel is easy to operate, especially with help of **SPTU**;
- **Intelligent interconnection (RS-485, CAN, Ethernet, BLE, 5G, HPLC, WIFI, NFC);**
- **High-precision measurement, power quality analysis;**
- Functions include: power distribution, short circuit protection, overload protection, grounding protection, undervoltage protection and disconnection etc.;
- Complete accessories: closing electromagnet, undervoltage release, shunt release, electric operating mechanism and auxiliary switch;
- Optional accessories: interphase partition, key plate, mechanical interlock;
- Circuit breaker health monitoring and diagnosis (contact temperature rise, intelligent accessories);
- TFT-LCD Controller, touch operation, graphical setting interface.

Smart ACB_Features

CHNT



Predictive Maintenance with **conditional monitoring**



Identify switch **topology** automatically, locating fault, fast recovery and reduced power failure time



Accurate measurement, **line loss** analysis, electricity stealing analysis, and optimization of three-phase imbalance



Support multiple communication methods: RS485/HPLC/BLE, etc., which can be flexibly configured to meet the needs of different communication schemes on site.



Comprehensive intelligent perception, remotely upgrade firmware, and reduce operation and maintenance costs



More friendly of human-computer interaction, easy for users to operate and consult

Pilot: ShangHai and WenZhou trialed run on customer side, and product works well.



No.	Field Location	Result
1	ShangHai	All the circuit breakers work well in field till now.
2	WenZhou	

Intelligent MCCB Features

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a

High precision measurement: current and voltage 0.5%, power 1%

b

Comprehensive intelligent perception: real-time uploading of equipment operation status, event information and power failure reporting

c

Multiple communication methods: RS485, HPLC carrier, BLE and other communication methods

e

Small size: the installation wiring size is same with the existing products

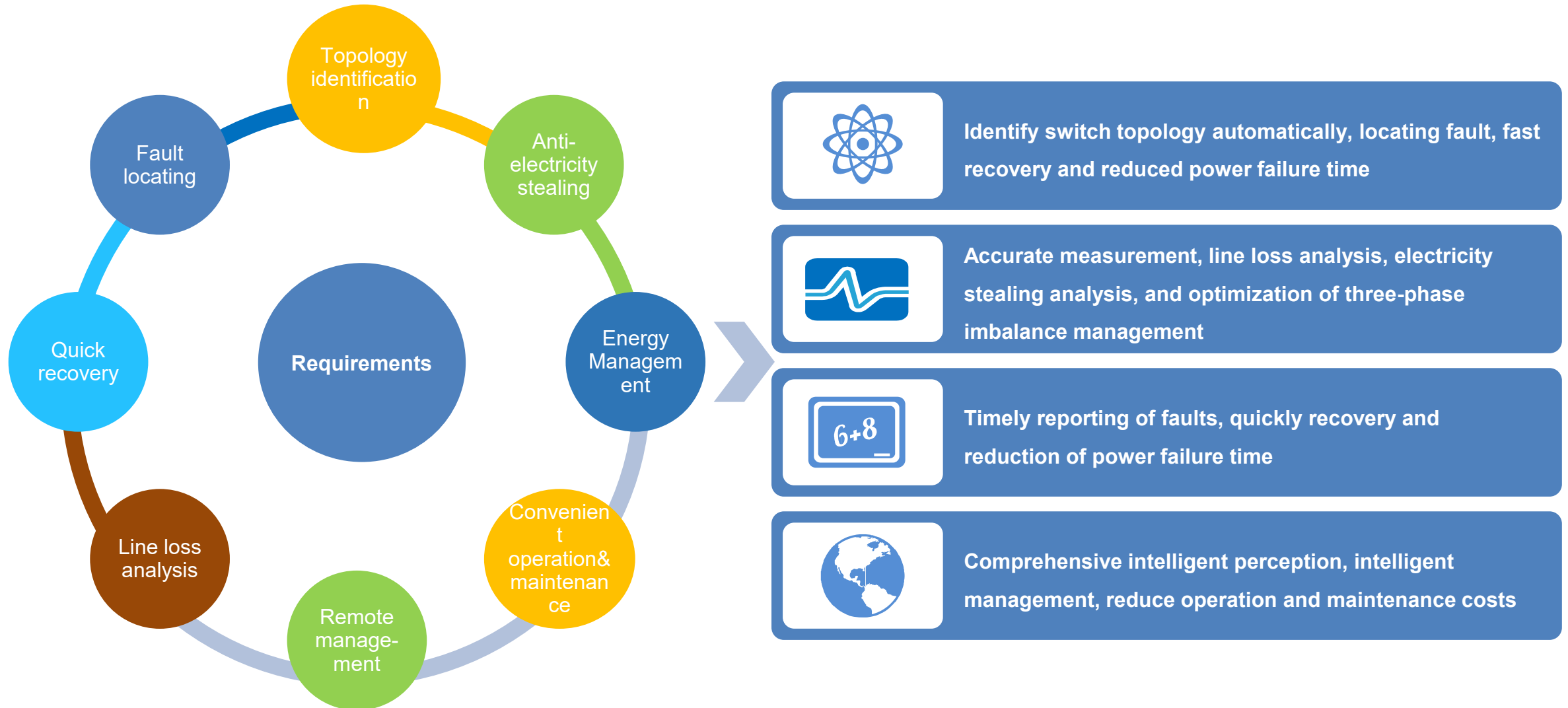
f

Topology identification: support to automatically generate topology diagram

Intelligent MCCB Requirements

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Pilot: Guangdong, Guangxi and other three provinces/cities trialed run at customer side, product works well.

Cooperate with the pilot installation and operation in Guangdong, Guangxi, Guizhou and other stations. Each breaker in the enclosure, branch box, metering box and other scenes in the substation area using CHINT intelligent MCCB products, and the data is uploaded to the China Southern Power Grid IoT platform to realize intelligent applications, such as real-time monitoring of circuit status, hierarchical management of line loss and remote query of fault information, to improve management ability and reliability.



No.	Field Location	Result
1	Guangdong: Zhaoqing, Jiangmen	All the circuit breakers work well in field till now. Total sales more than 20,000.
2	Guangdong: Heyuan, Yangjiang	
3	Guangxi: Qinzhou, Hechi	
4	Guizhou: Zunyi	

Intelligent measurement switch Features

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With HMI type

W/O HMI type

a

High-precision measurement, high-precision clock function, support the analysis of ammeter misalignment and abnormal power consumption

b

Modular design, measurement unit supports hot plug, automatically adapt to different shell frames

c

Detachment-proof lead seal of the measuring unit to prevent unauthorized replacement of the measuring unit

d

active, reactive energy pulse output, accuracy can be verified.

e

Support automatic generation of topology identification diagram

f

OLED: supporting data viewing and parameter setting

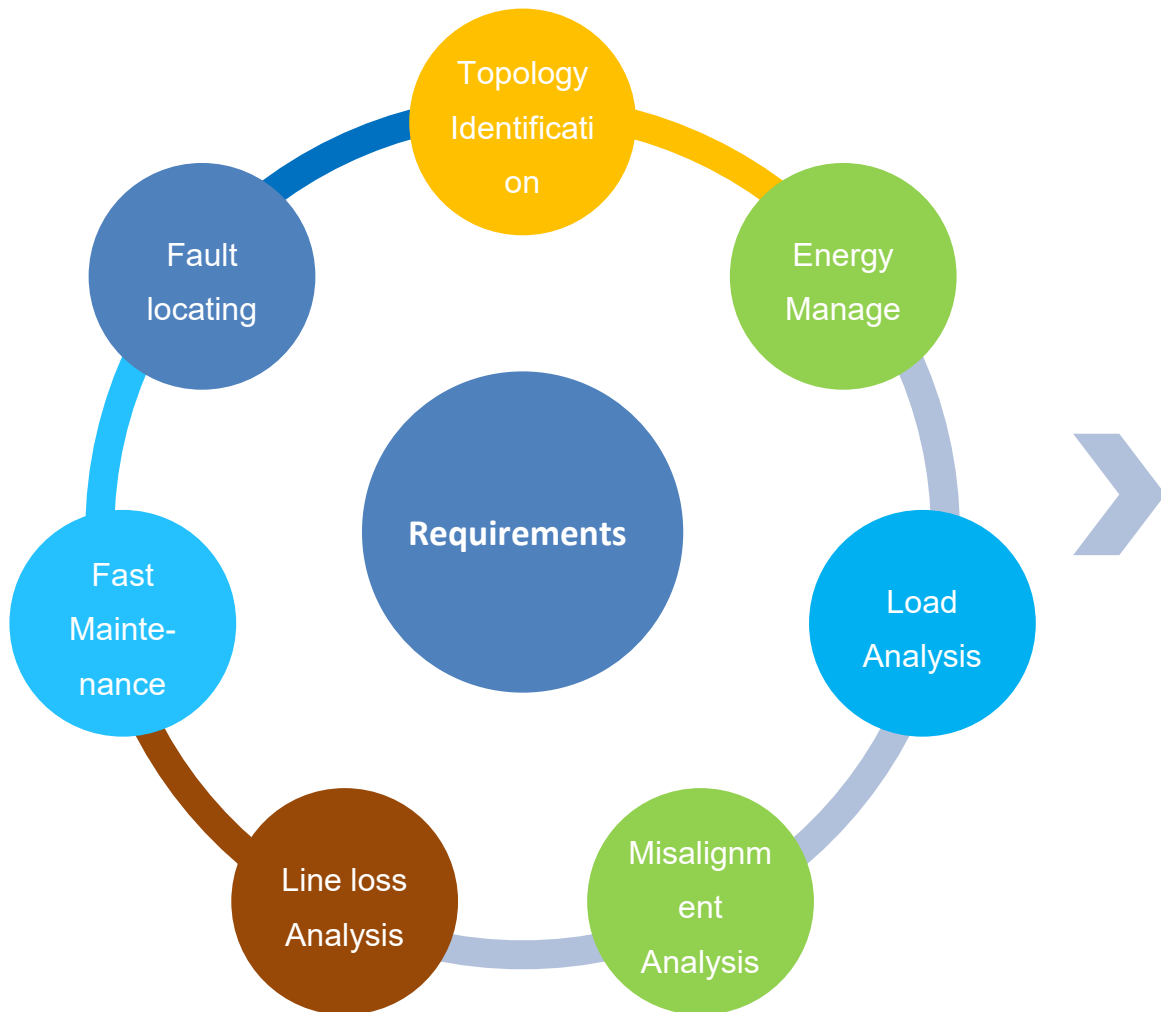
g

Support multiple communication methods: RS485/HPLC/BLE, etc., which can be flexibly configured to meet multi-scene needs on site.

Intelligent measurement switch Requirements

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Using pluggable modules to achieve power failure detection, recognition, and reading, ensuring uninterrupted power consumption



The inaccuracy justify of the ampere meter is realized through statistical analysis based on the whole system, the master meter, the house meter and the measurement circuit breaker



The measuring accuracy is up to 0.5S class, and the data is same as meter. Also the collected data can be compared and analyzed with the corresponding meter under defined accuracy condition.



Report power failure and recovery information from meter cabinet, and is convenient to check the power failure and recovery status of meter system.



Provide effective support for distribution area about line loss and power load analysis.

Intelligent measurement switch Application

CHNT

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Pilot: 5 provinces in China, such as Beijing and Shandong, trialed run at customer side, and products work well.



No.	Field Location	Result
1	Beijing: Tongzhou District, Yanqing District	By now, those field tests in power grid work well. The measurement data is correct. Daily increment data analysis works well. Total sales of 50000+.
2	Tianjin: Hebei District	
3	Shandong: Yantai, Dezhou	
4	Henan: Simulation Station Area of EPRI	
5	Chongqing: Changshou District	

Intelligent leakage reclosing MCCB Features

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a

Comprehensive protection: provide protection about residual current, three-stage, grounding, over-voltage and under-voltage, phase loss, neutral break etc., fully guarantee safety of electricity system.

b

All electric parameters can be measured: it can measure current, voltage, power, electric quantity, etc., providing convenience for load monitoring

c

High-precision measurement: current, voltage, power $\pm 0.5\%$, 0.5S class measuring , providing accuracy data support for line loss analysis

d

Multiple communication methods: RS485, HPLC (optional), BLE (optional) communication mode to realize the interconnection and intercommunication of intelligent nodes in the substation area.

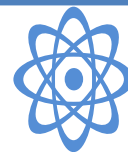
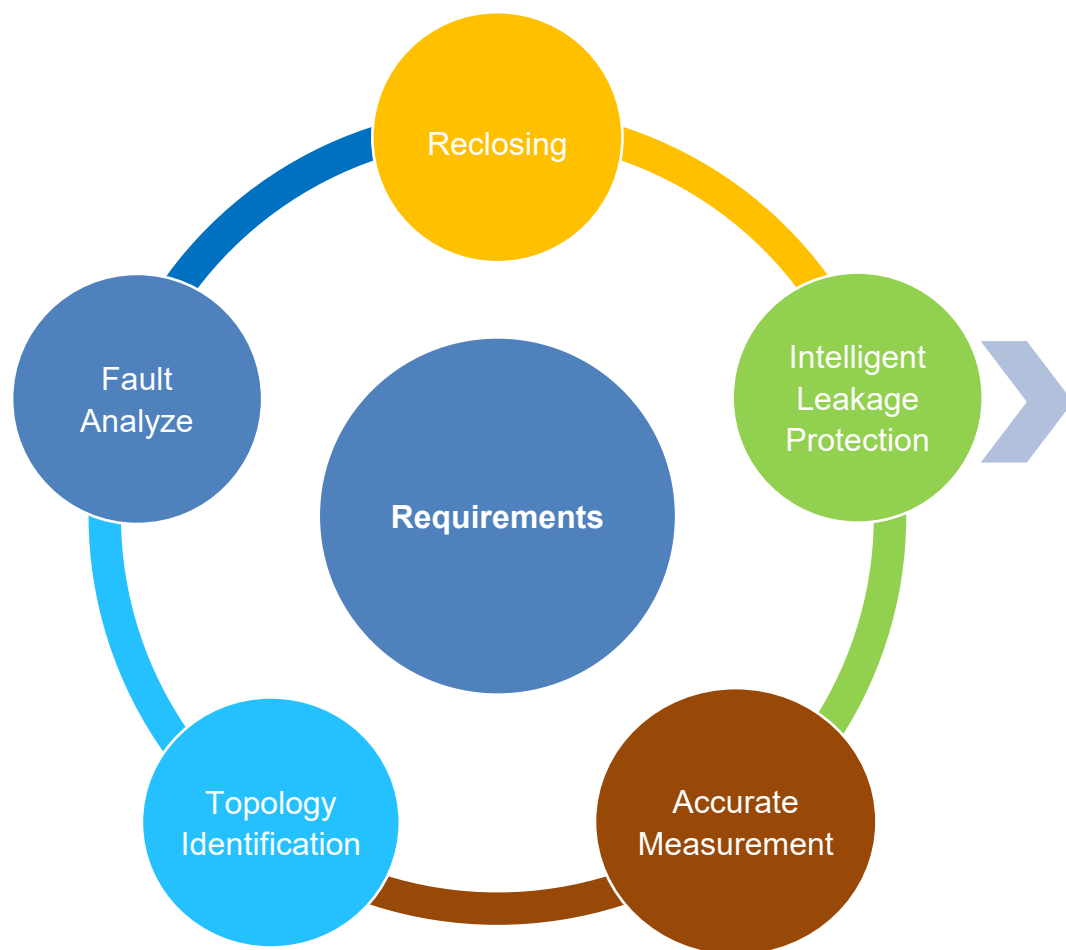
e

Innovative design: to provide support for expanding application scenarios and optimizing application value by improving HMI, thinner design, controller power optimization and other innovative designs.

Intelligent leakage reclosing MCCB Requirements

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The reclosing function after residual current, overvoltage, undervoltage and phase loss fault tripping can quickly restore power supply and shorten the repair time.



Residual current protection, fault spot automatic tracking and positioning. If special wave or sudden change happens, breaker protects the system with the continuity of power supply.



Accurate measurement, providing data support for line loss lean management and electricity stealing analysis



It supports automatic identification of topology, and can realize accurate positioning of fault points and rapid maintenance



It provides fault analyze and justify, and clear causes of power failures, then restore power supply quickly, to reduce power failure time, at the same time reports fault information to upstream system.

Intelligent photovoltaic grid-connected breaker Features

CHNT

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NM3LFC Photovoltaic Grid-connected Circuit Breaker



a

- **Passive islanding protection** Realize automatic opening when mains power failure, multi-feature fusion algorithm is adopted to realize millisecond change response to distortion of voltage, frequency, and phase.

b

- **Power quality detection:** the frequency refresh rate can reach 5ms, and the harmonic refresh rate can reach 50ms, which can realize real-time monitoring and rapid response

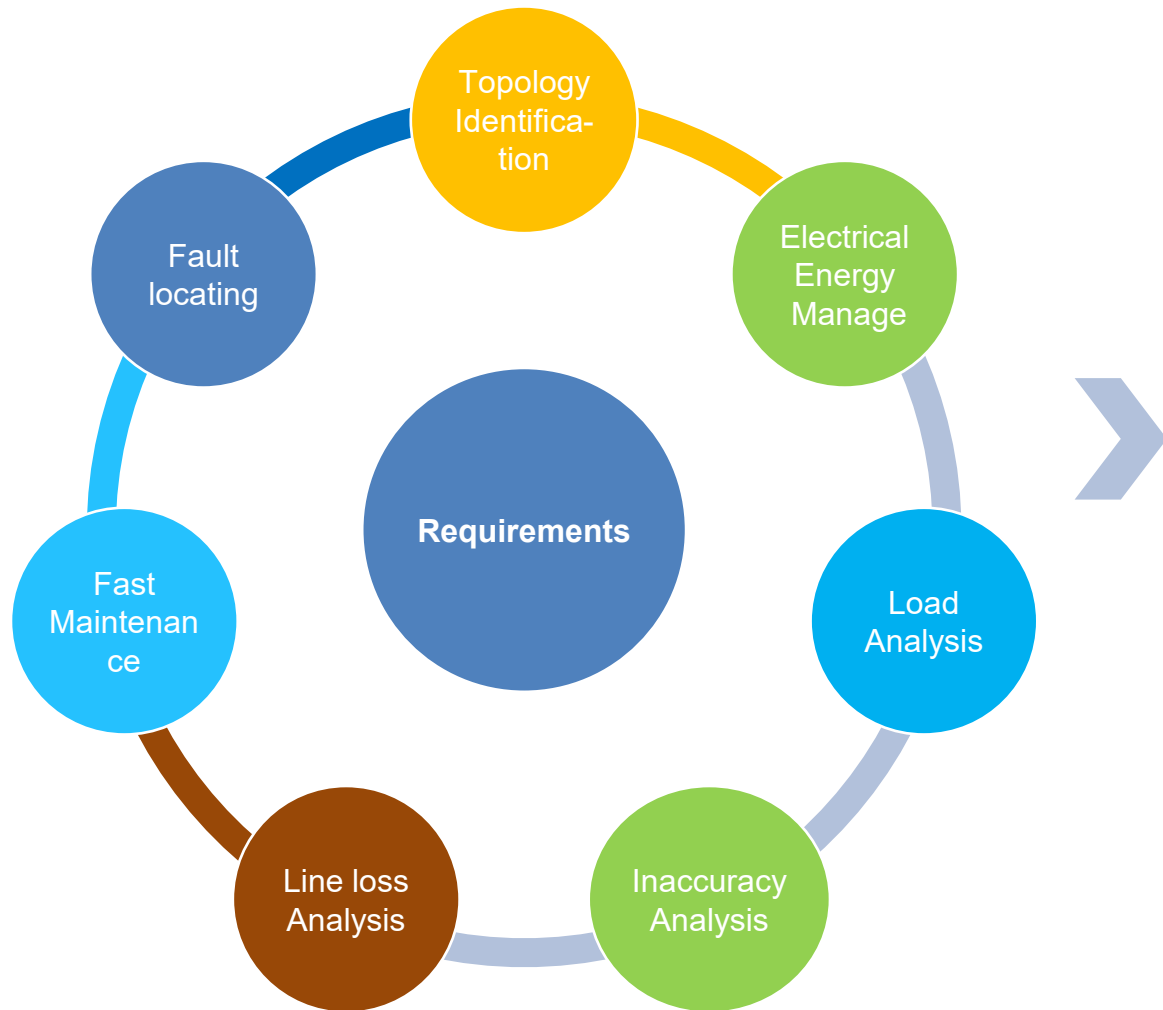
c

- **Terminal temperature measurement & protection function:** Close cooperation with the mechanical design, the ceramic independent temperature measurement sensor is designed to prevent the fire caused by the overtemperature of the terminal

Intelligent photovoltaic grid-connected breaker Requirements

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The inaccuracy justify of ampere meter is realized through statistical analysis based on the acquisition system, the master meter, the house meter and the measurement circuit breaker.



The measuring accuracy is up to 0.5S class, and data reported is same as the meter displays, so the collected data can be compared and analyzed with the corresponding meter under defined accuracy condition.



Report power failure and recovery information from meter cabinet, and convenient to check the power failure and recovery status for meter system.



Provide effective support for distribution area about line loss and power load analysis.

PV Breakers Interface Protection Compliance Review

No	Interface Protection(IP) Items	CHINT PV Breakers		
		ACB (NA8-S type)	MCCB (NM3LFC)	MCB (NBP-GVS)
1	Undervoltage	●	●	●
2	Overvoltage	●	●	●
3	Overfrequency	●	●	●
4	Underfrequency	●	●	●
5	Loss Of Mains(Anti-island)	●	●	●
6	Instantaneous Overcurrent	●	●	●
7	IDMT Overcurrent	●	●	●
8	Earth Fault Current	●	●	O

“●” – Complied, “O” – not complied currently

Intelligent photovoltaic grid-connected breaker Application

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Pilot: Shandong, Henan, Liaoning 3 provinces and cities trialed run at customer side, and products work well.



No.	Field Location	Result
1	Shandong: Jinan, Qingdao, Yantai, Weihai, Binzhou and other cities	Manual opening and closing works well.
2	Henan: EPRI Simulation Platform	Automatic opening and closing works well.
3	Liaoning: Liaoyang, Chaoyang	Remotely opening and closing works well. Total sales of 800,000+.

Smart meter external MCB Features

CHNT

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a

Through BLE, remote opening/closing, closing permission and other operations can be realized. Avoid wiring and paring fault, save labor cost.

b

Through BLE communication, reading the user event record status. Active reporting during breaker status changes, manual/automatic mode optional

c

According to DL/T698 protocol, it is supported by ammeter, terminal and master station, and can be well extended

d

BLE automatic pairing, software encryption, current **fingerprint technology**, providing quicker paired and better security.

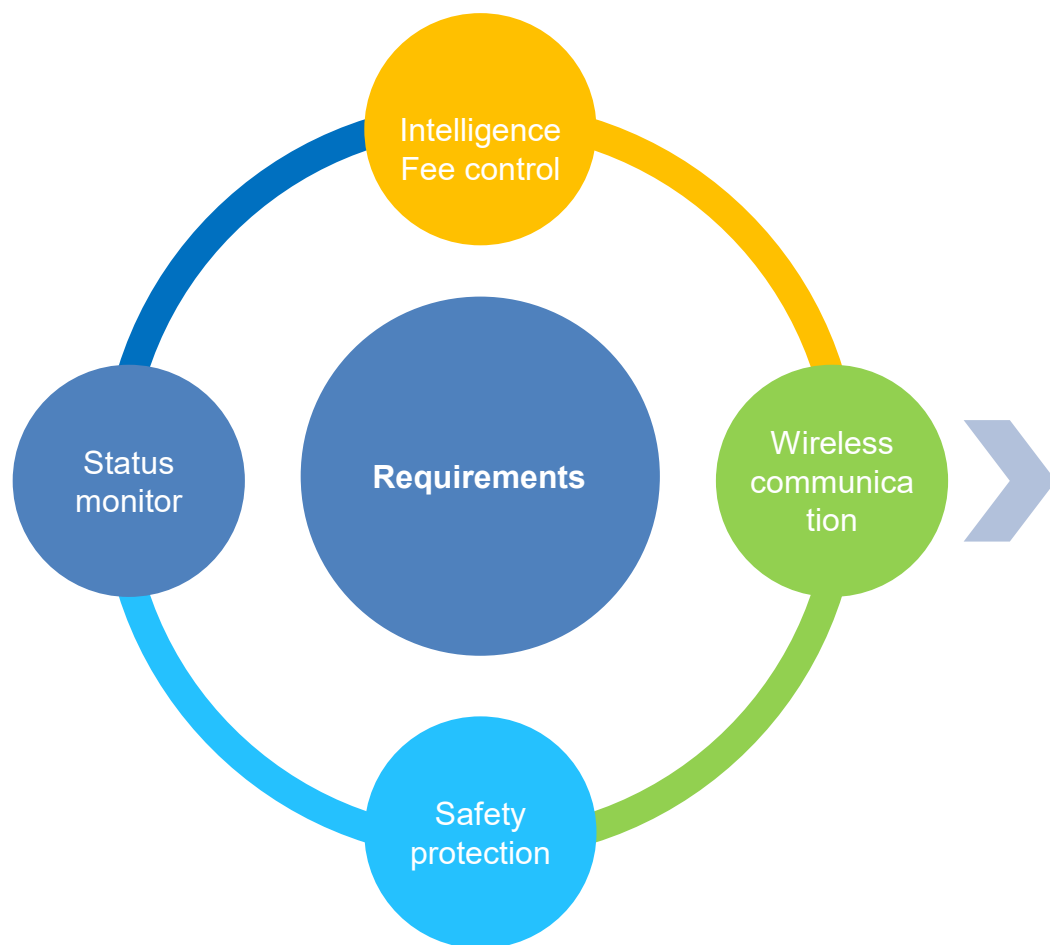
e

No human supervising on-site needed

Smart meter external MCB Requirements

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Intelligent management through BLE to cut off line during arrear with electricity, power on after recharge



Real-time monitoring and accurate justify switch opening and closing status, power failure and restoration events and fault status



Providing BLE method to connect to smart IoT meters for save labor cost.



BLE charge control, anti-damage, software encryption, power distribution protection, guarantee electricity safety



Automatic pairing using current fingerprint technology to avoid manual errors, allowing for pairing when the device is not powered on

Smart meter external MCB Application

Pilot: Jiangsu, Henan and other 4 provinces and cities trialed run at customer side, product works well.

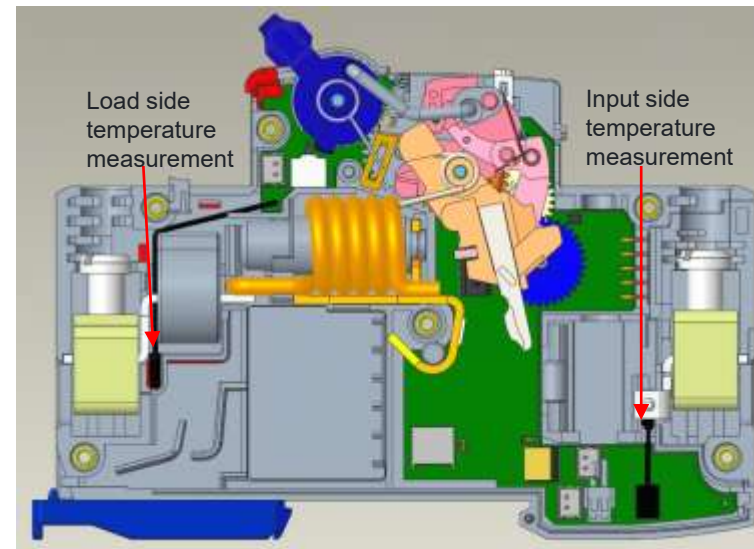


No.	Field Location	Result
1	Jiangsu: Huaian City	Remotely charge control opening and closing Upload the operating status of circuit breaker When the manual mode is opened, the background cannot be closed remotely. Total sales more than 10,000pcs.
2	Shandong: Linyi, Jining	
3	Henan: Simulation Station Area of EPRI	
4	Chongqing: Hechuan Power Supply Bureau	

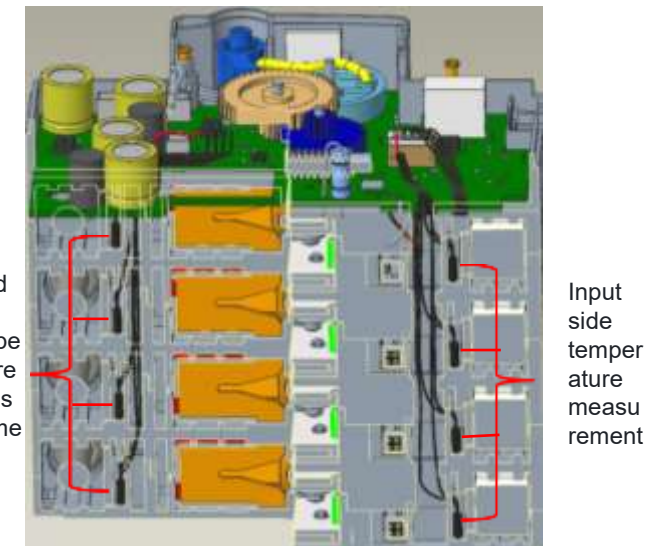
- **Core technology**
- Temperature sensing technology
- For the first time in the industry, the temperature measurement function of the incoming line end and the load end is realized in the miniature circuit breaker to monitor the wiring health of Uimp protection technology.
- In the smart MCB, the design of temperature probe and wire routing, meet the requirements of temperature measurement accuracy of terminal, and pass 9.8kV Uimp test.
- **Key results**
- Meet the requirements of HIKVISION and fire protection



NB2 Series 1P- > 3PN



Temperature measurement of 1p product terminal



3PN product terminal temperature measurement

Pilot: 3provinces and cities in China, such as ShangHai,Xian and WenZhou, trialed run at customer side, and products work well.



No.	Field Location	Result
1	ShangHai	All the circuit breakers work well in field till now. Total sales more than 20,000.
2	WenZhou	

Intelligent combined LV shunt capacitor

CHNT

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a

The product has small size, low power consumption, fast response, better compensation effect, and all round protection.

b

Through cable connection, it installs easily and expands conveniently.

c

Free of secondary wiring and no need maintenance, it saves labor costs for end user.

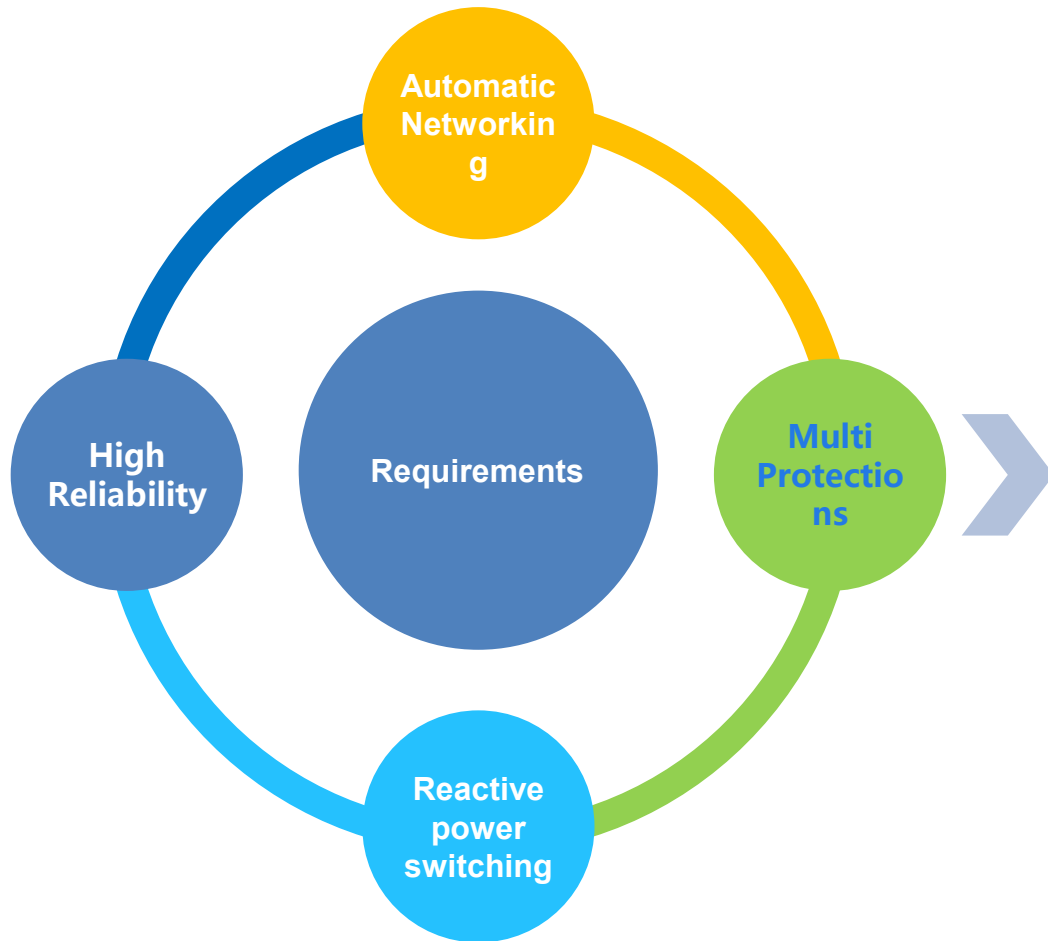
d

With reactive power switching, the capacitor's compensation effect is better, and it can bring the customer with a better use experience.

Intelligent combined LV shunt capacitor

CHNT

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It supports three-phase co-compensation, single-phase compensation, hybrid compensation, phase-to-phase compensation, and can provide customers with different requirements .



The product allows 1 million times of switching, with high reliability, and can reduce customer operation and maintenance cost.



It has the ability of automatic networking, and no secondary wiring, so it can reduce customer labor cost.



It can self-diagnose and self-recover, and it has over temperature, and over current protection, so it can guarantee product life span and use safety.

Intelligent combined LV shunt capacitor Application

CHNT

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Pilot: ShangHai and WenZhou are trialed run at customer side, product works well.



No.	Field Location	Result
1	ShangHai	All the circuit breakers work well in field till now. Total sales more than 10,000pcs.
2	WenZhou	

PV Rapid shutdown

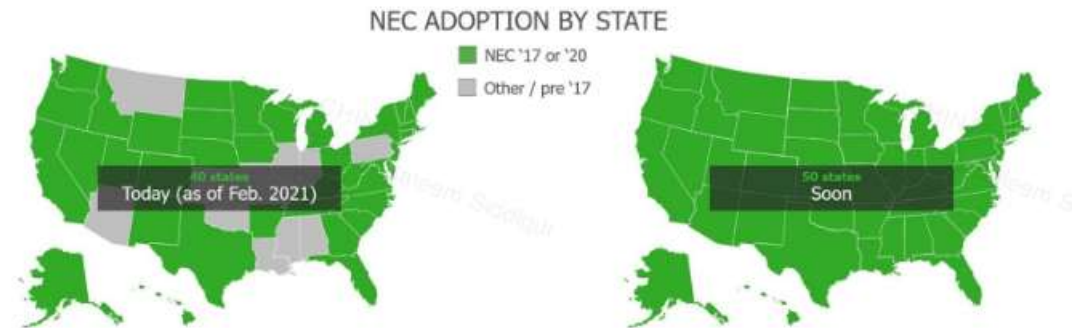
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- There is a major security risk for photovoltaic systems that is **Rescue risk**.
- The traditional series system is prone to **DC ARC Phenomena**, and **DC High Voltage** leading to fire hazards.
- When a fire accident occurs the PV module continues to generate electricity the high voltage is constantly present.
- This situation prevents firefighters to approach the scene and control the fire from a safe distance.

It can be dangerous for firefighters if safety measures are not followed.



PV Rapid shutdown

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Product Features:

- Unique product structure design, excellent heat dissipation function.
- Secure and reliable, The maximum input current can reach 20A.
- Easy for maintenance, When the circuit is broken, the output voltage is 1V.

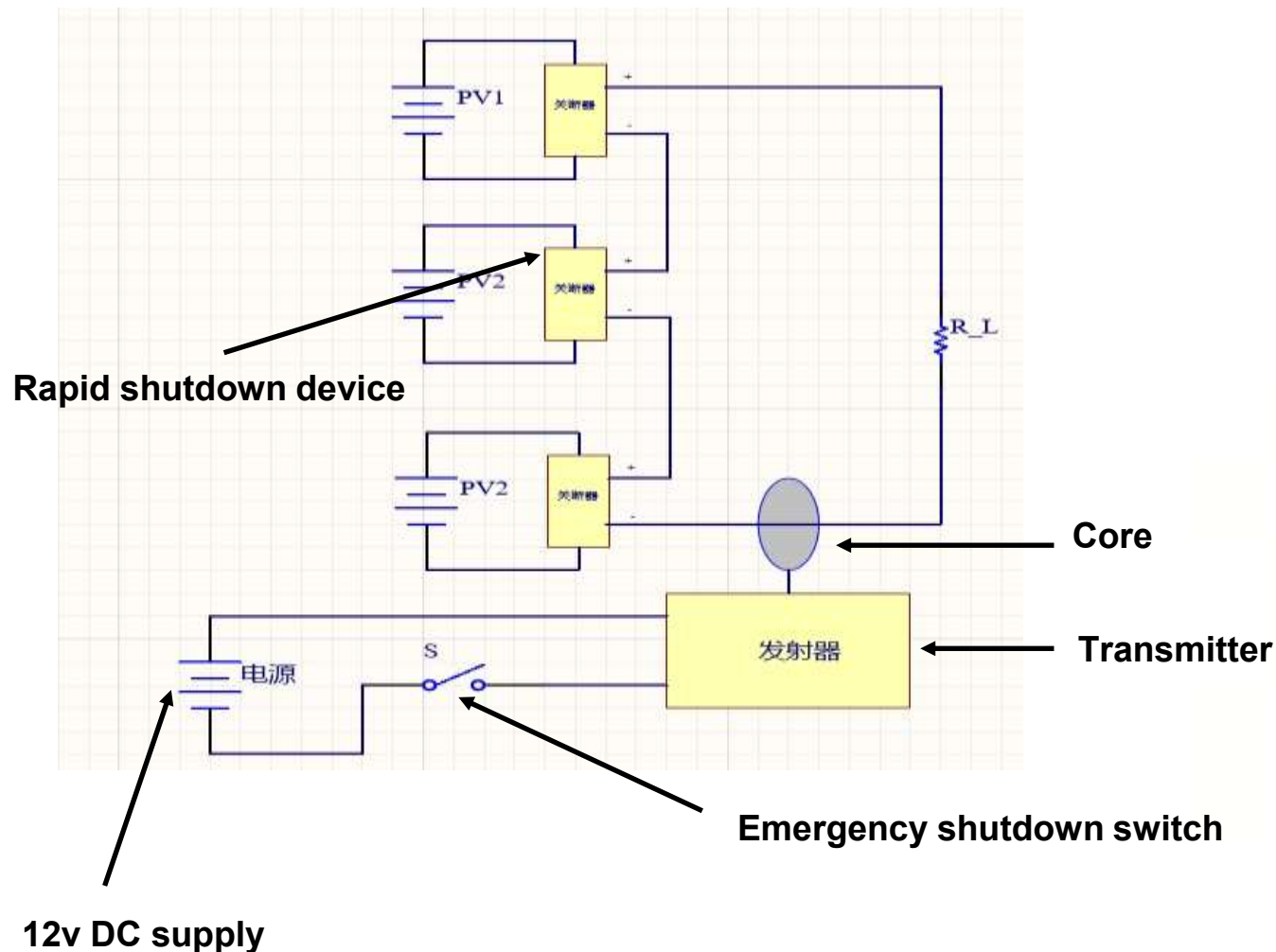
- Rapid shutdown is an electrical safety requirement applies to solar PV systems.
- PV rapid shutdown quickly disconnect and isolate output of each panel of photovoltaic (PV) system.
- Ensures that DC voltage from solar panels is rapidly reduced to safety levels.
- Required by safety regulations such as **NEC 690.12**.

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Tamir Z. Mubiani 704d 2146

PV Rapid shutdown

Product application diagram



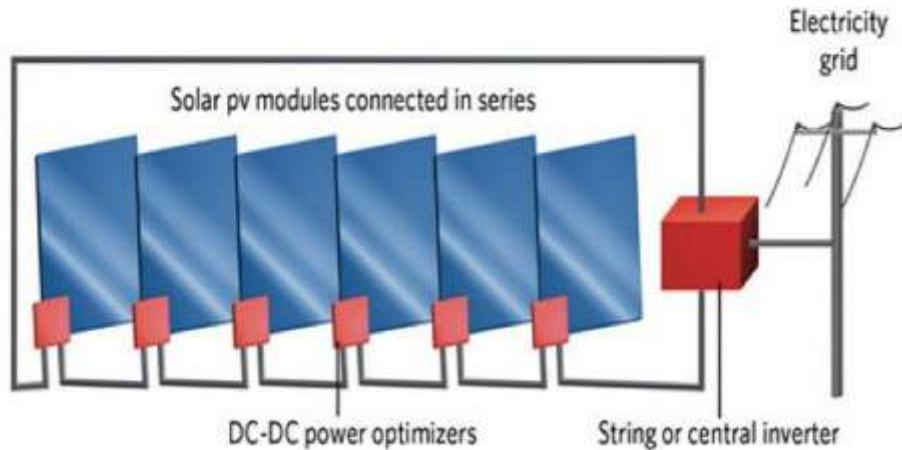
- **Rapid Shutdown Controller:** This is the central device that initiates the rapid shutdown sequence. It receives signals from various triggers, such as manual switches or sensors, to deactivate the PV system quickly.
- **Rapid Shutdown Device (RSD):** This device is installed on the back of each solar module. When it stops receiving signal from transmitter it rapidly stops the flow of electricity from the solar panels.
- The voltage levels come down to safe levels very quickly.

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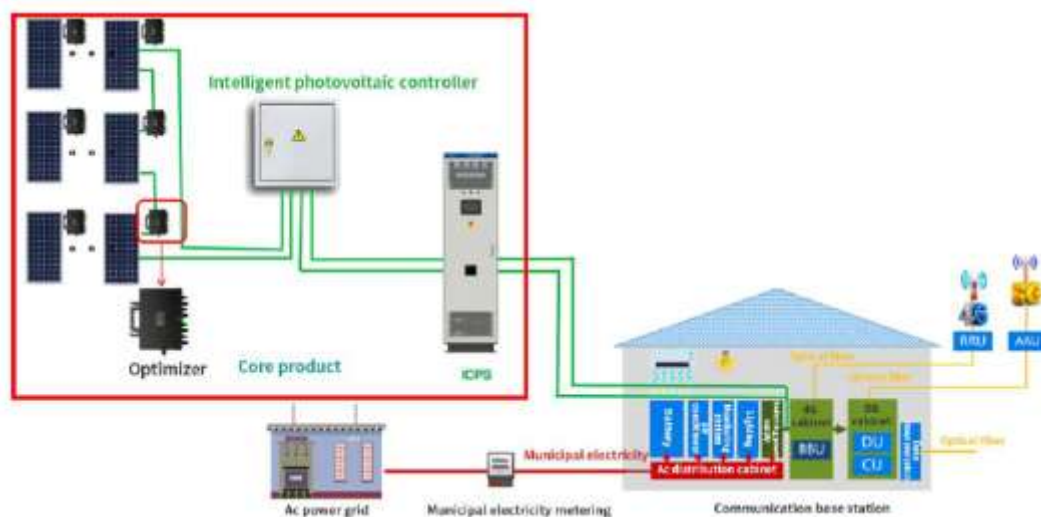
PV Power Optimizer-Modular MPPT

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- ❑ A **DC-DC converter** installed at the back of each PV module.
- ❑ Performs **Maximum Power Point Tracking (MPPT)** for each panel.
- ❑ Interfaces with central or string inverters.
- ❑ May include communication and safety features.



2146

Tamil

EV Charger –Sequence Charging

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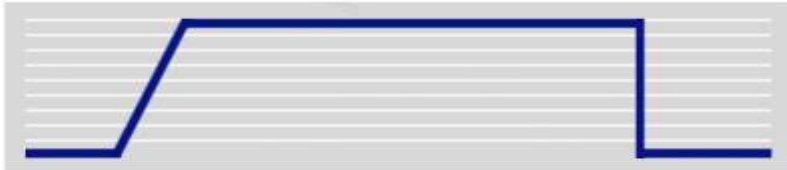


- ☐ A surge in EV charging can place a significant load on the local sub station.
- ☐ If the grid is not equipped to handle the additional load, this can result in voltage fluctuations or even power outages.
- ☐ Charging many EVs simultaneously can lead to a degradation of system.

新能源

EV Charger –Sequence Charging

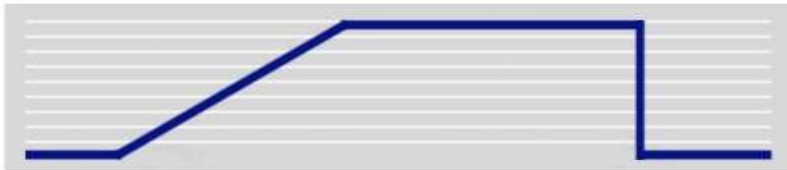
Charging
Pile #1



Charging
Pile #2



Charging
Pile #3

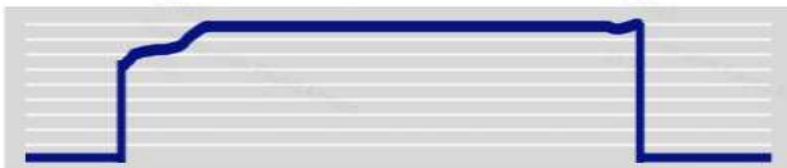


⋮

Charging
Pile #N



Total



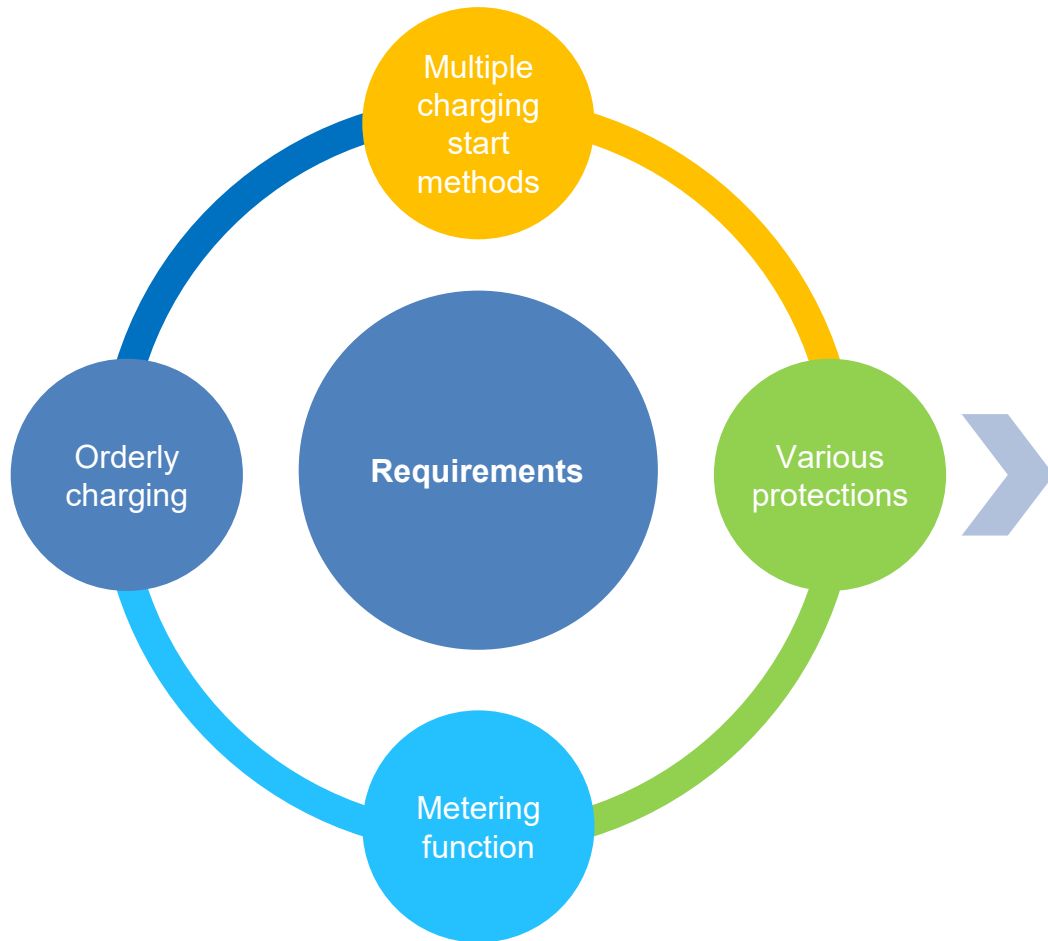
- ☐ Sequence charging is the solution to reduce the pressure on sub stations.
- ☐ Sequence charging can detect the high demand.
- ☐ It reduce the charging capacity which results in reduction of the load on sub station.
- ☐ When multiple charging piles are present, **modulation** can refer to how the system coordinates their usage to ensure balanced load distribution across the grid.

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Charging pile Requirements

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Product includes multiple charging methods and is suitable for personal and commercial purpose



Product has a high-precision, and can use as commercial billing or personal metering



Various protection functions



Charging current adjustable used for orderly charging



Data security: all data transmit with encrypted

LV DC Application technology

CHNT

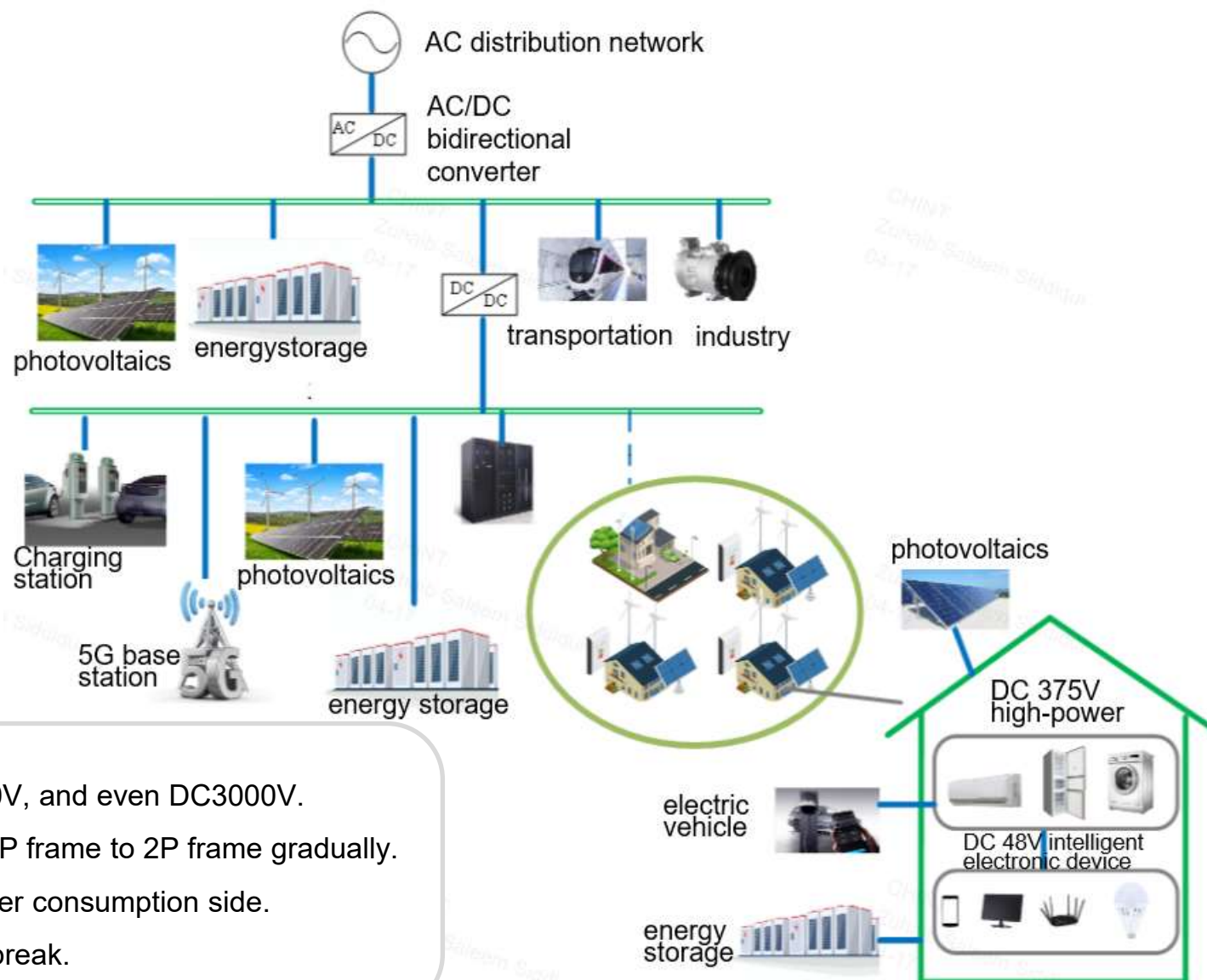
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Decreasing dependence on oil resources has been one of the goals of the government since the 1970s. In the manufacturing industry, promote the localization of renewable energy research and development manufacturing and industrial equipment manufacturing, and achieve the initial goal of 9.5 gigawatts of renewable energy generation.

The new power system with renewable energy as the main body has put forward higher requirements for the application and development direction of low-voltage DC products.

Application trends

- A. Rated voltage: higher voltage, tends to be DC2000V, DC2500V, and even DC3000V.
- B. Structural trends: smaller size, changing from the traditional 4P frame to 2P frame gradually.
- C. End users: Development from DC distribution side to DC power consumption side.
- D. Safe and reliable: providing leakage protection by DC circuit break.



DC ACB

1) DC1500V DC frame circuit breaker

Used in data centers, 5G communication facilities, and DC main distribution systems

- Housing grade: 2500 A, 4000 A
- Rated voltage: DC750/1000/1500V
- $I_{cu}=I_{cs}=I_{cw}=50KA @4000A$
- **$I_{cu}=I_{cs}=I_{cw}=45KA @2500A$**
- Electrical life: up to 2000 times



2) ASD16 DC Disconnect (UL & IEC)

Applied to domestic and foreign photovoltaic, energy storage and other high voltage DC systems

- Housing rating: 1600 A
- Rated current: 600 ~ 1600 A
- Rated voltage: DC750/1000/1500V
- **$I_{cw} = 75KA 50ms, 150KA 20ms$ Industry Highest**
- The first product in China and abroad to meet UL and IEC standard certification



3) DC1500V DC frame disconnect

Centralize inverter apply to photovoltaic and energy storage

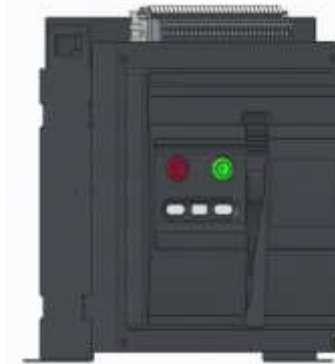
- Housing grade: 2500 A, 4000 A
- Rated voltage: DC750/1000/1500V
- I_{cw} up to 100KA @ 4000A
- **Electrical endurance up to 3000 times @ DC1500V 2500A Industry's highest**



4) DC1500V DC disconnect 2P

Applied to DC side of high voltage inverter of domestic/foreign photovoltaic system and energy storage system

- Housing grade: 2500A
- Rated current: 600 ~ 2500 A
- Rated voltage: DC750/1000/1500V
- $I_{cw}=150KA 200ms$
- **2P compact structure, minimum volume**



DC MCCB

1) DC1500V DC MCCB 2P

High-voltage box applied to energy storage system

- Housing grade: 400A, 630A
- Rated current: 250 ~ 630A
- Rated voltage: 1500V
- $I_{cu}=I_{cs}=15kA$ (10ms)
- Electrical endurance: 1500 times
- **CCC, CE, TUV certification**



2) DC disconnect

Applied to domestic and foreign photovoltaic systems, energy storage systems and other high-voltage DC systems

- Housing grade: 320A, 630A
- Rated current: 125 ~ 630A
- Rated voltage: DC1500V
- Service category: DC-21B, DC-PV2, $I_{cw} = 8kA$ @ 1s (320 housing) $I_{cw} = 10kA$ @ 1s (630 housing)
- **It is the only one in the industry that meets three directions operation(front, side and back), and the operation direction is the same.**



3) DC1500V DC double cut-off MCCB

Centralize inverter for photovoltaic and energy storage

- Housing grade: 250A, 400A, 630A
- Rated current: 100 ~ 630A
- Rated voltage: 1000 V/1500V
- **$I_{cu}=I_{cs}= 25kA$ @5ms**
- Electrical endurance: 1500 times
- **CCC, CE, UL certification**



4) DC1500V single-break MCCB 2P

Centralize inverter apply to photovoltaic and energy storage

- Housing grade: 630A
- Rated current: 250 ~ 630A
- Rated voltage: 1500V
- $I_{cu}=I_{cs}= 15kA$ @10ms
- $I_{cu}=I_{cs}= 25kA$ @8ms
- Electrical endurance: 1000 times
- **CCC, CE, TUV certification**



4) Introduction of New Products and Technologies

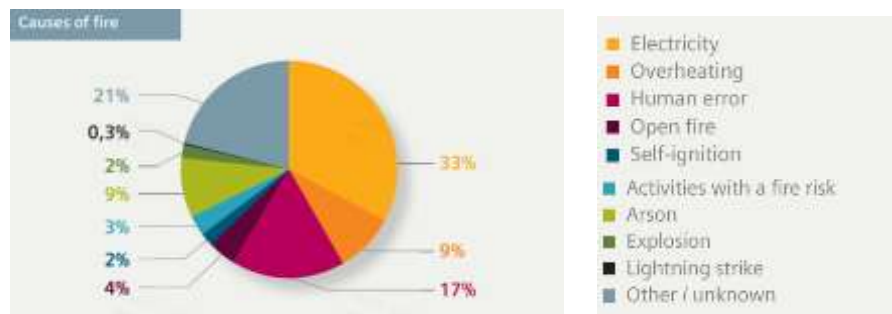
Arc Fault Detection technology

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• Research background

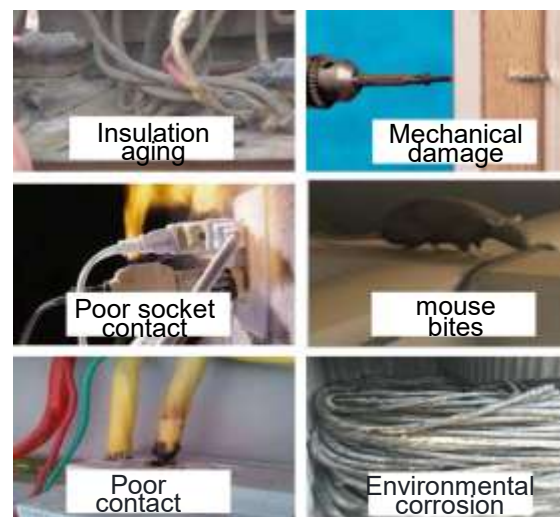
- Arc fault is usually caused by insulation damage and poor connection, with symptom hiding, continuous smoldering, and finally be on fire.
- Arc fault is mainly divided into series arc, parallel arc and grounding arc, in which the series arc current is lower than the steady-state current, and the conventional over-current and leakage protection devices can not detect it.



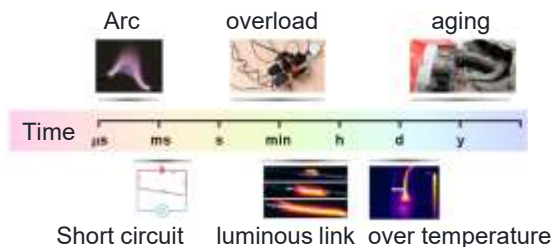
Causes of fire in Germany in 2014



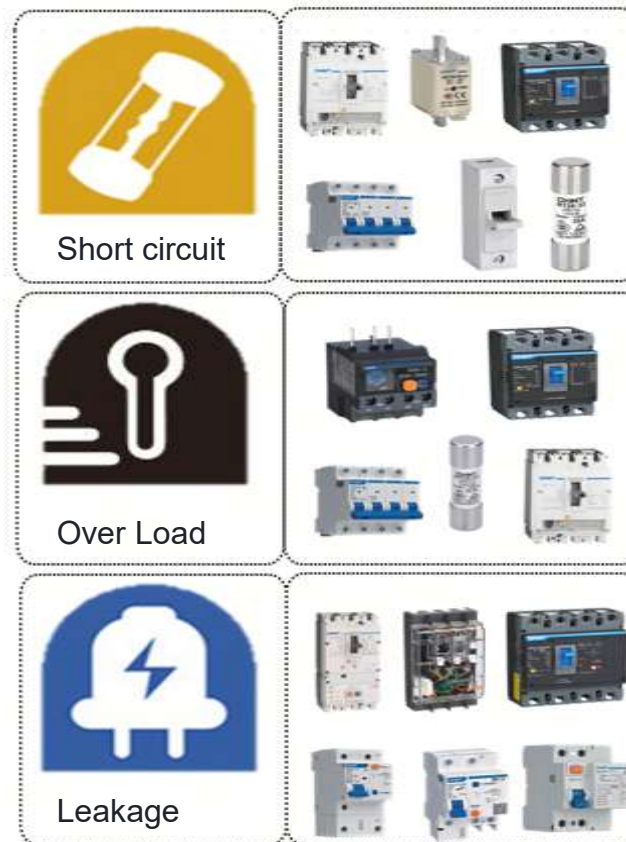
French cathedral (Notre-Dame de Paris)



Arc generation causes



Arc generation spans multiple time scales such as aging, over-temperature, etc.



Conventional distribution protection devices

Arc Fault Detection technology

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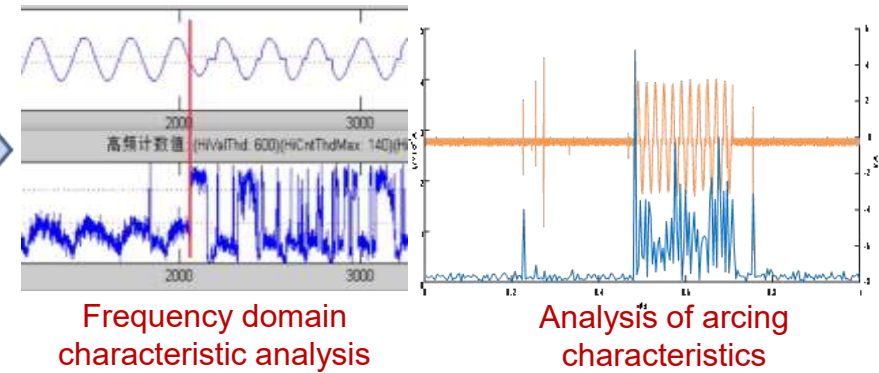
• Key technologies _ Arc fault detection technology based on multi-feature recognition

- Avoids the influence of arc-like load, and the identification accuracy is intuitive and reliable, which is according to the typical characteristics of arc current in time and frequency domain, the corresponding algorithm is developed to identify the arc.
- Arc recognition rate of existing scene reaches more than 85% by the classification and identification of arc features and local rapid data processing, which promotes the rapid iteration of products and the expansion of application scenarios.

AC Arc fault detection

- The quality of AC arc detection products in the market is uneven, and the rate of missing and false alarm is high;
- The application environment of AC arc is complex, and the influence of arc-like load needs to be avoided, so it is difficult to detect.

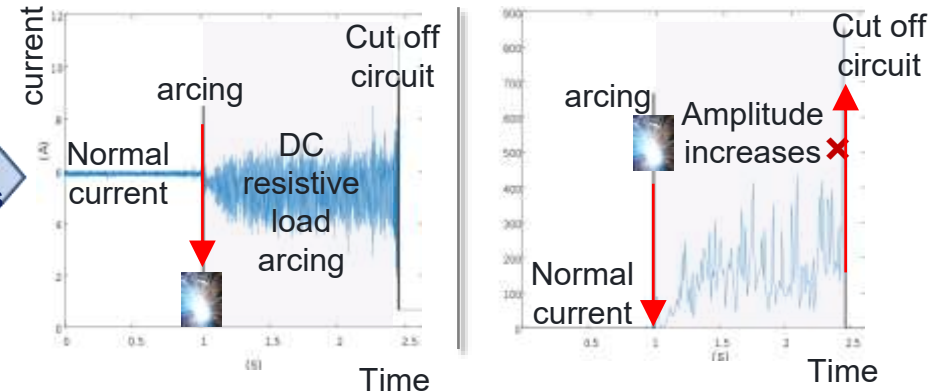
Feature analysis



DC Arc fault detection

- DC arc discharge is stable, and the frequency domain signal characteristics below 10 kHz are obvious.
- The load of photovoltaic system is single, and the arc detection is mainly to avoid the interference of inverter.
- Arc identification is combined with photovoltaic fast shutdown to achieve component-level shutdown under fault conditions.

Feature analysis



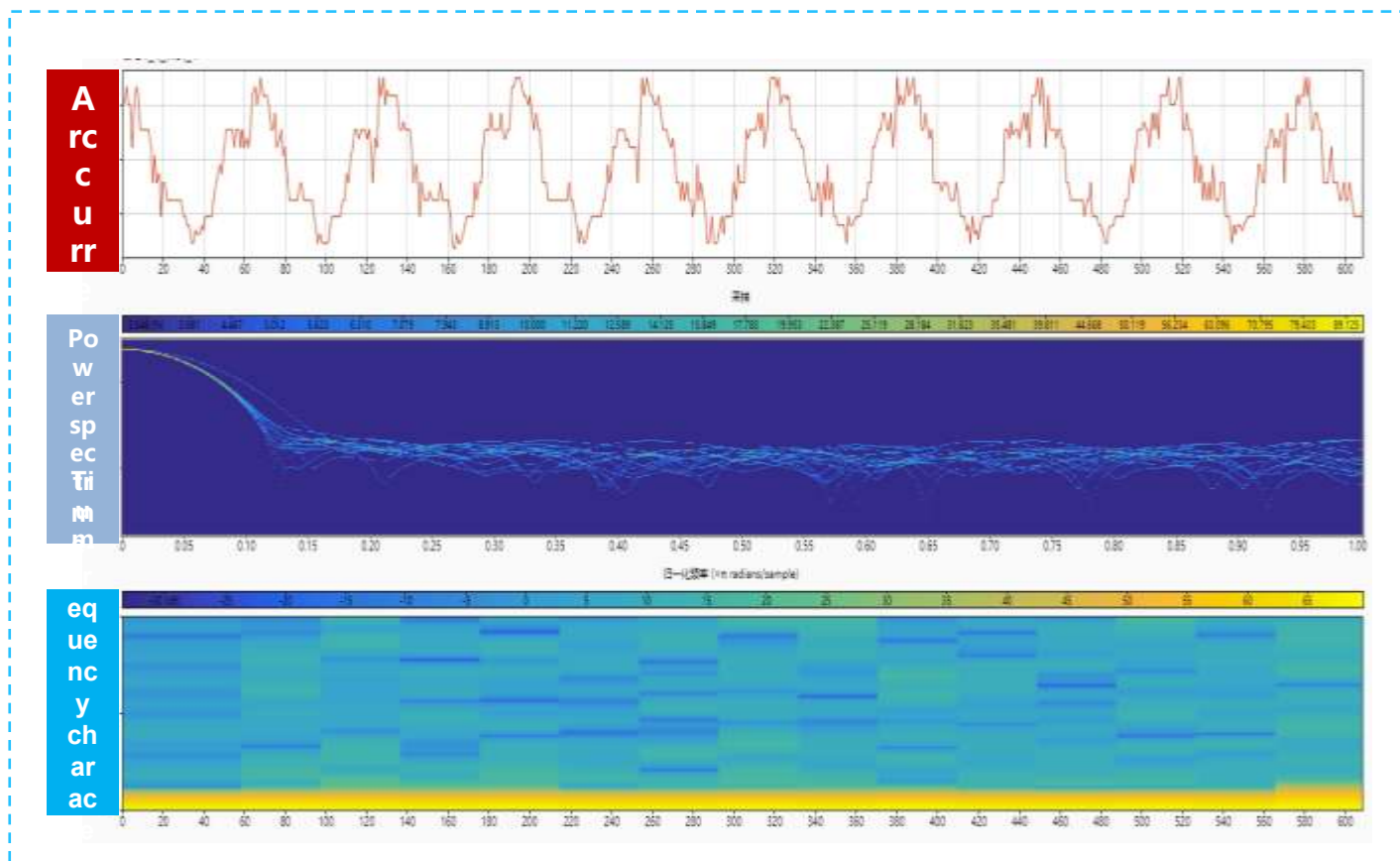
Arc Fault Detection technology

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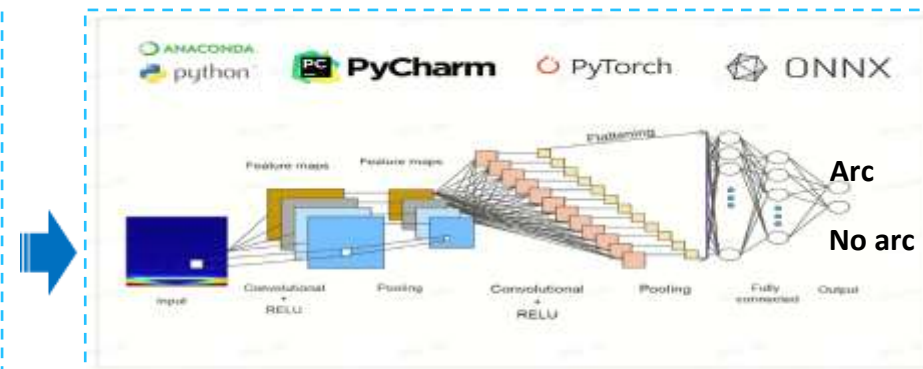
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• Key technologies _ Arc detection technology based on machine learning

- Base on feature reconstruction visualization which is achieved by FFT, DWT, multi-dimensional data identification and other technologies, arc model training is accelerated by AI algorithm.
- The lightweight model based on machine learning is superior to the traditional arc recognition scheme under laboratory conditions, with a recognition accuracy of 90% +, and the product reproducibility needs to be further improved



Feature reconstruction visualization



Fault arc model training

批次	SVM	BP	Alex Net	*Net Module
accuracy				
1	75.3	83.2	87.9	93.1
2	78.1	86.4	91.2	95.7
3	74.2	84.3	88.5	93.2
4	69.7	81.5	86.9	91.3
5	77.4	82.1	85.8	93.5
6	80.3	87.3	84.2	95.2
7	76.5	85.5	89.6	96.6

Comparison of recognition rate of different algorithm

Arc Fault Detection technology

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• R&D achievements

- AFDD products have been sold more than 20,000 pcs since launched in 2022, which has been continuously upgraded with customer feedback.
- In view of the rapid iteration demand of products, intelligent AFDD product development and cloud edge collaboration are carried out to build a power safety system with fine management.
- DC arc detection has expanded from a single field of photovoltaic grid connection to DC electricity field, but DC electricity field hasn't been defined details in detection standards and implementation methods yet.



NB3LE-AFD (Production)



NB4LE-AFD (Production)



NB2LE-80ZT intelligent AFDD (pre-research)

NILM - Non-intrusive Load Monitoring:

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- Monitor consumption for each load connected to mains
- Identify irregular loads
- Ensure safety and legal compliance



Electrical bicycle indoor charging in residential units



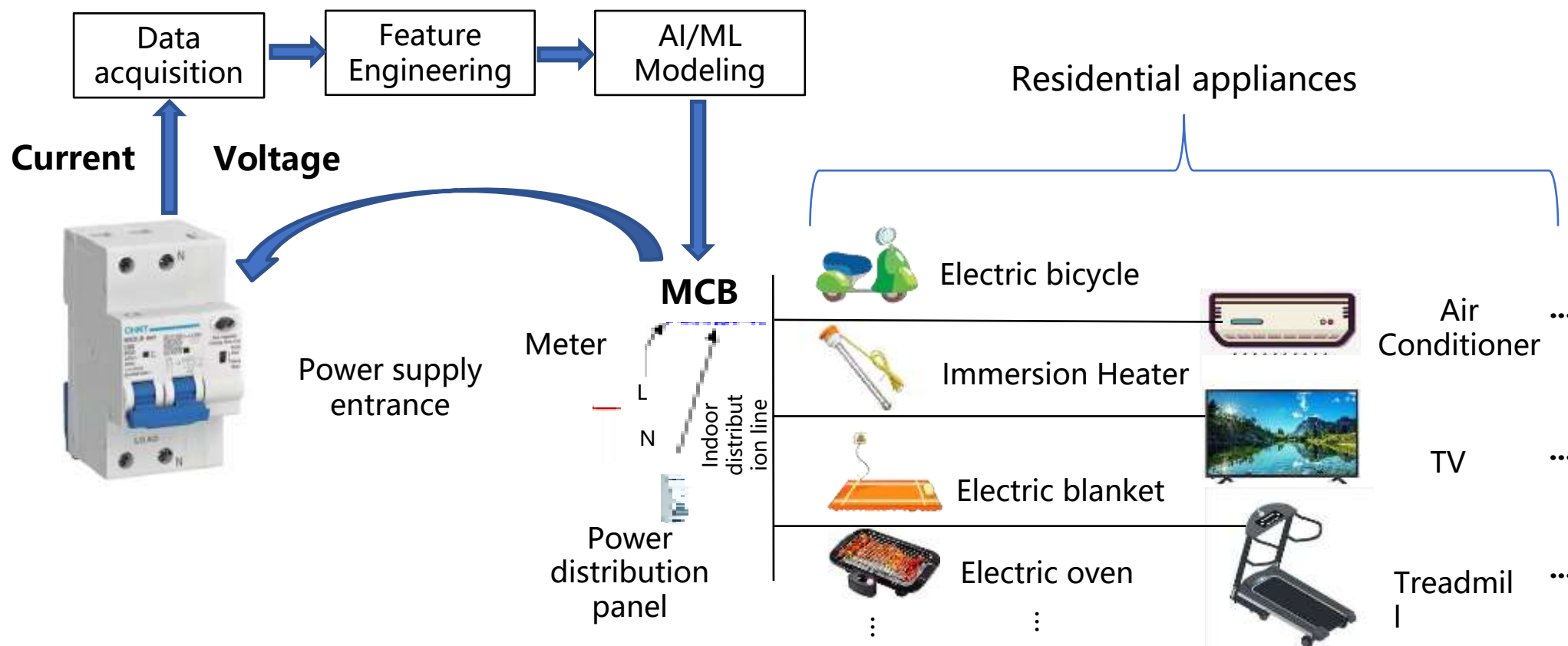
High-power prohibited appliances on campus

NILM - Non-intrusive Load Monitoring:

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- Data extracted from one single metering / signal acquisition point
- AI inference at the edge / meter level device
- No extra meters or extra feedback nodes
- Integrated with energy management system



NILM - Non-intrusive Load Monitoring:

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- Real-time alarm and statistics
- Load Identification (ie Illegal heavy equipment/EV charger)
- Sophisticated feature recognition / statistics per feature



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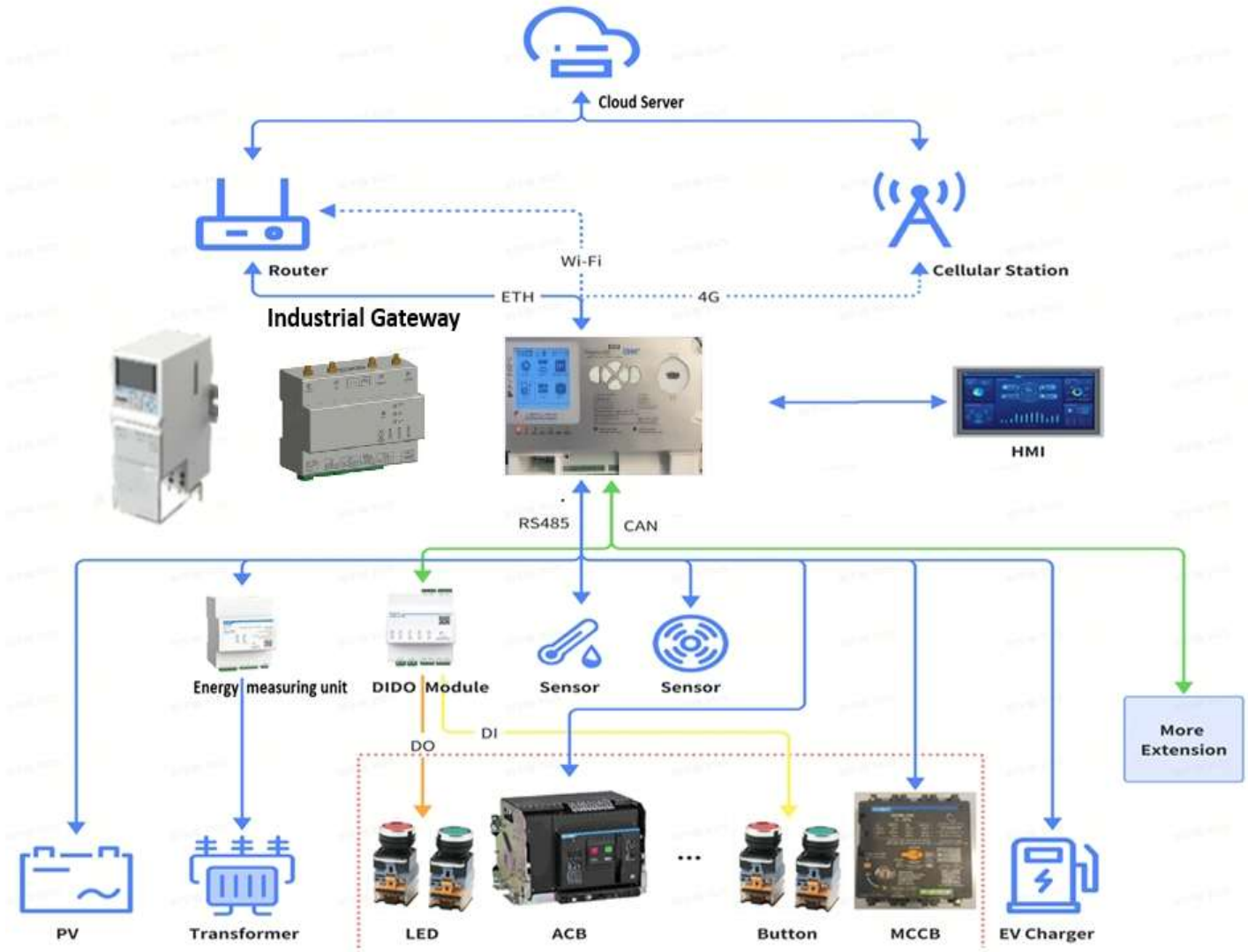


CHINT Industrial Gateway system solutions

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- CHINT's years of experience in the automation, distribution and transmission industry can help customers fulfil their requirements of innovative technologies like Industrial Gateway.
- According to the field environment, LoRa, Wi-Fi, BLE, Wi-Sun, CAN, RS485, 4G LTE module can be provided.
- Industrial Gateway and measurement unit can be self power supplied no matter it is single phase, 3 phase or 3 phase 4 wire.
- We can provide the best and most cost-effective solutions to suit different application scenarios.



CHINT Industrial Gateway system Features:

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The industrial gateway is designed to be modular, allowing it to be integrated with other units such as DIDO modules, communication modules, power quality modules, sensor modules, and more, based on the client's requirements.

✓ Modular Design



The system also provides an interface between modules, allowing communication through various technologies such as CAN, USB, and RS-485. Additionally, it supports sharing the power supply to simplify connections and usage.

✓ Interface between Modules



The Industrial Gateway has an HMI accessory to support local data monitoring without opening the cabinet.

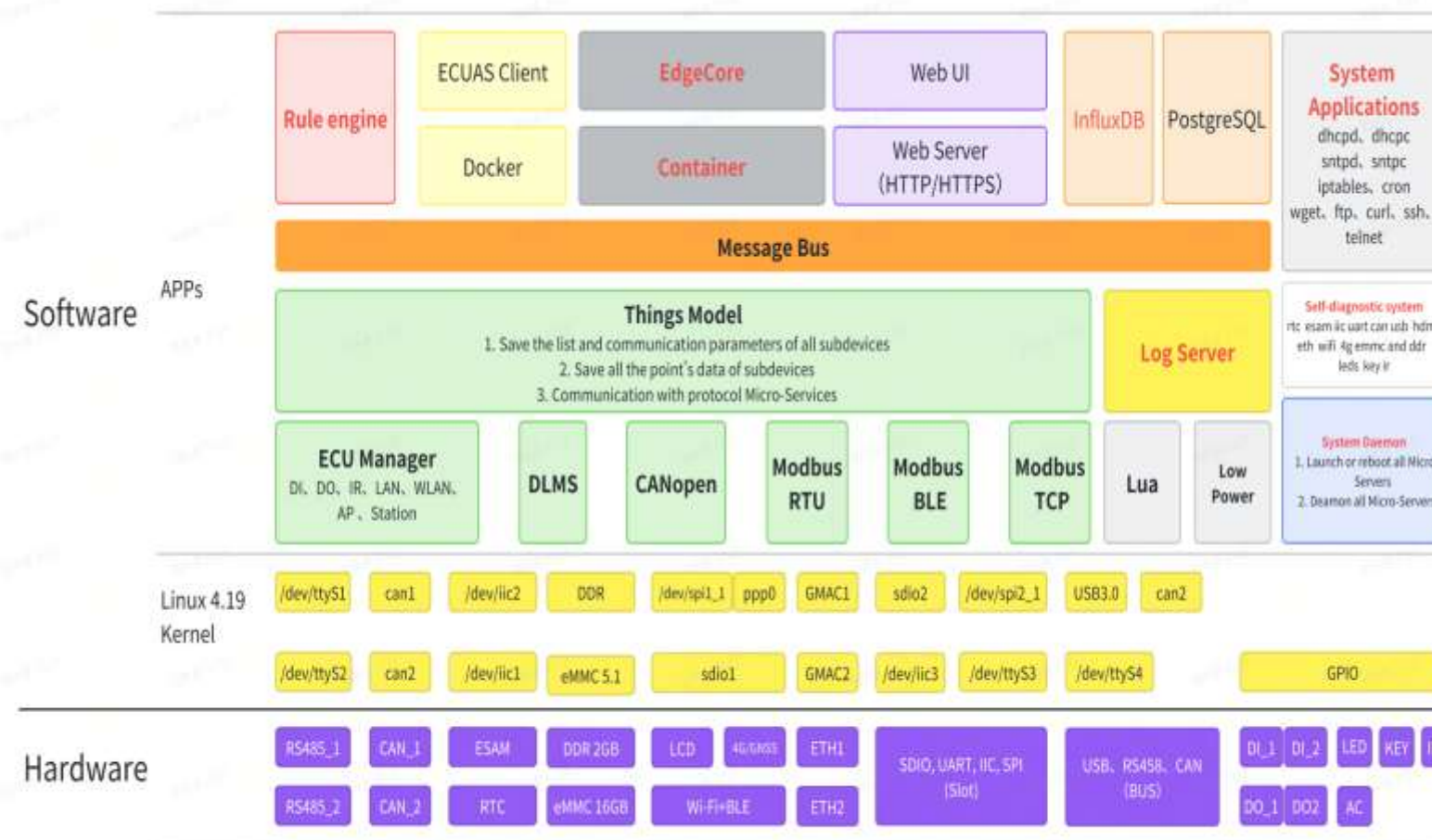
✓ HMI Accessory



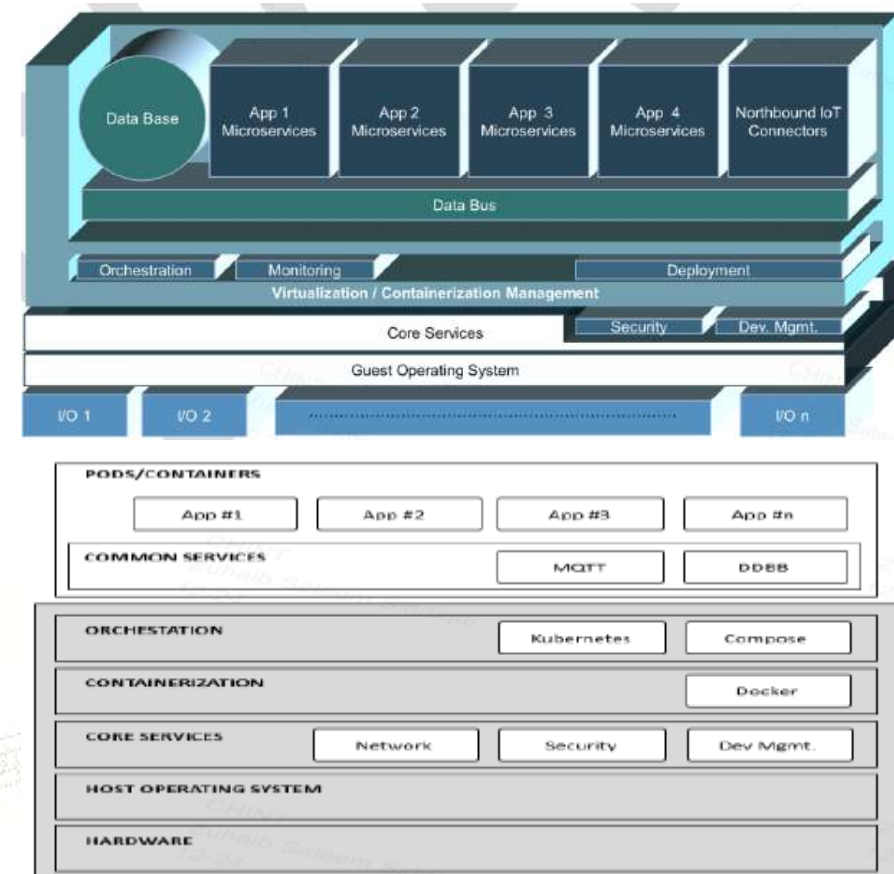
CHINT Industrial Gateway system Features:

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System Architecture



SEC specification

CHINT Industrial Gateway has already considered all the important features like system Daemon, Docker and Rule Engine

The docker will help us flexibility deploy the application

The influxDB help us fast save the sensor's data and data interaction between the app of docker

EMS - Energy Management System:

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IoT & AI-powered energy management system

Industrial, commercial, and residential applications.

- **Predictive maintenance** to prevent failures and minimize downtime
- **Energy Allowance Planning** utilizing history of energy consumption together with external data (i.e.: Predict weather for PV resource planning)
- **Real-time insights** – Monitor and analyze energy data
- **Reduce energy** consumption by up to 20%
- **AI-driven** optimization to automate efficiency and cost saving
- **Cut waste** and lower carbon footprint



Monitor Power Quality



Asset Management



Efficiency Reporting



AI - Analysis & Prediction

EmpowerX – Infrastructure for Seamless Integration:

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- **Comprehensive Framework** that fuses smart CHINT technologies together
- **Connectivity** across all levels – consumer/establishment to cloud to Vendor
- **Efficient Data Usage** of CHINT products, mixing OEM know-how with in-house software design
- **On-Premise and Cloud** deployment options for maximum flexibility



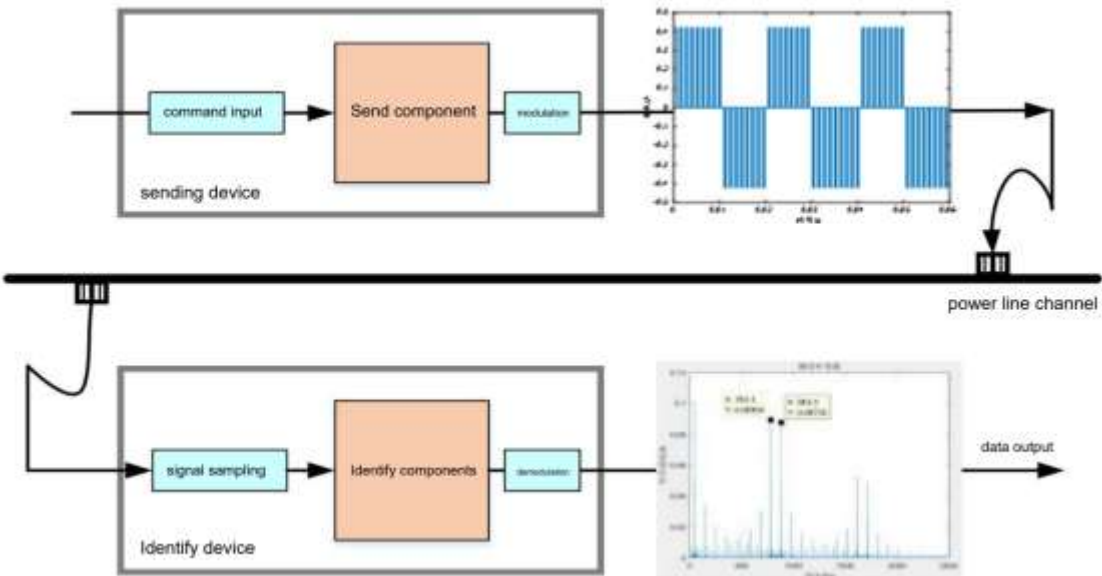
THANKS

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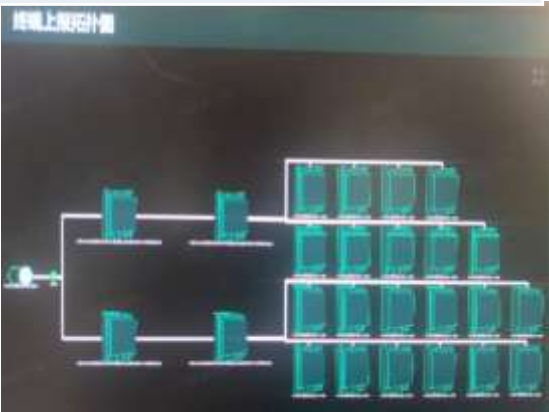
Topology Automatically Identification

- Based on characteristic current injection technology, different topology of power distribution areas can be identified, analyzed and recorded at the server, generating a virtual topology map among all devices. Combined with proactive fault reporting, it enables global equipment to sense and locate fault quickly.

Mainstream technology	Principle	Identification time	Advantage/Disadvantage
Characteristic current injection technology	Additional hardware module needed, Based on event recording	Short (hour level)	Can generate topology during line debugging, even without load; Need to install additional hardware module fore each, which will increase the cost; Can not be applied in the photovoltaic (current is bidirectional); Would interfere in the power grid (0.5A current).
Pure software implementation	Based on algorithm	Long	No need to add hardware, low cost, and no impact on the power grid. It requires a load to be recognized; once the load is replaced, it needs to be relearned and recognized;



1. Smart terminal unifies the station node time through the carrier and clears old records.
2. Branch nodes send feedback with characteristic current signal after receiving the trigger command;
3. Upper branch nodes identify characteristic current, then record and upload identification time;
4. Smart terminal generates a virtual topology diagram through analyzing the time between sending and receiving information from each node, then reports it to the main station.

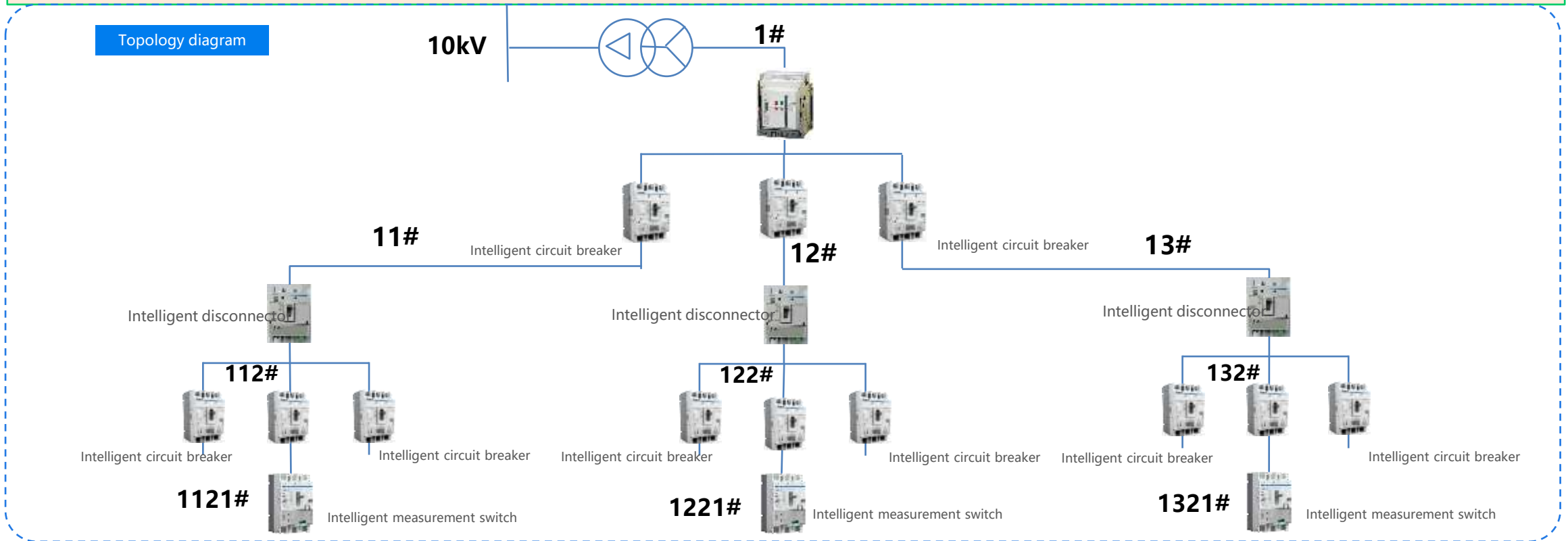


At present, field trials have been carried out in Shandong, Xiong'an, Hu'nan, Zhejiang and other places in China, and the identification rate of the trials can reach more than 90%;

Architecture block diagram of topology identification based on characteristic current

Intelligent topology identification

The built-in HPLC module of the nodes (intelligent switch) supports the topology automatically identification function, which improves the operation and maintenance management level of the distribution area.

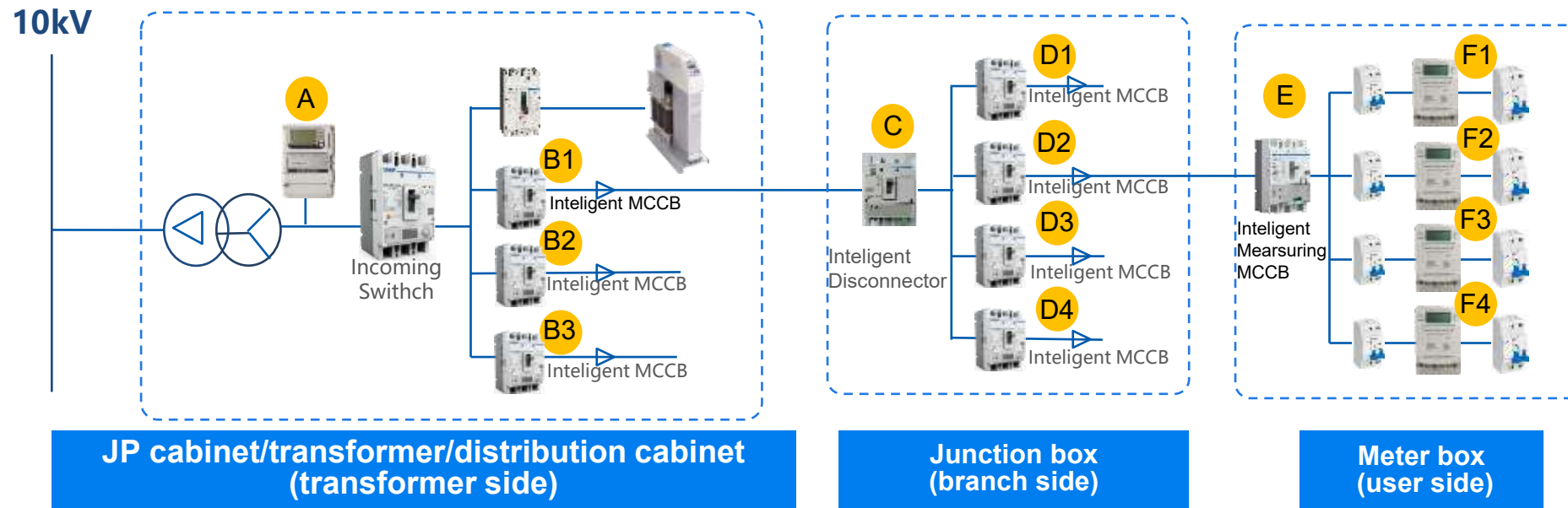


With the help of topology automation identification, we can find and locate the fault quickly and accurately, including the fault substation area, the fault line, the fault circuit breaker etc., the line emergency repair efficiency would be improved, and the maintenance time would be reduced.

Line Loss and misalignment Analysis

Line loss and misalignment analysis introduction

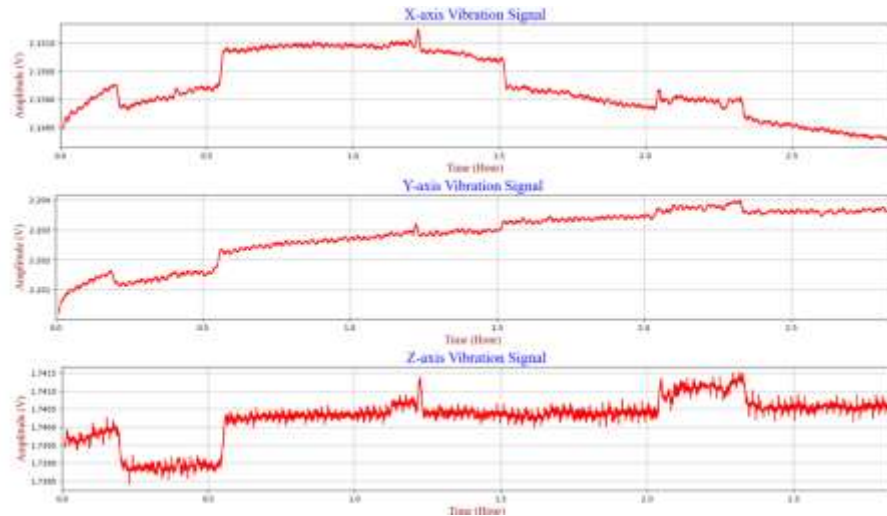
1. Compare the difference between the value of A main meter's electricity and the summation of each F* meter (F1...F4) in the area, calculate and analyze the line loss rate of the area, and check whether the line loss of the area is within a reasonable range.
2. If the line loss rate in the substation area exceeds a reasonable range, further calculations can be performed based on the metering data of the intelligent switch (B) on the transformer side, the intelligent switches in the branch box (C), and the main switch (E) before the meter, to analyze the circuit or specific location where the line loss occurs, and arrange staff to conduct on-site inspections.
3. If the power of the main meter (A) in the substation is significantly different from the power of the intelligent switch (B) on the transformer side, it can be inferred that meter A is inaccurate or existing electricity theft or leakage current or others failure.



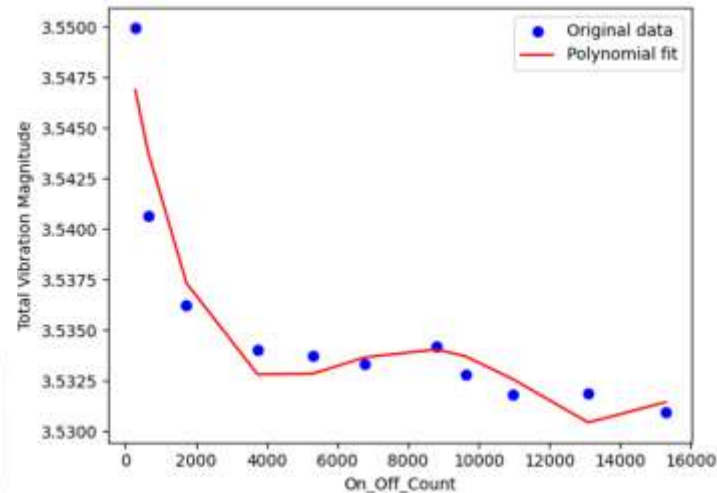
Condition Monitoring:

- Smart Predictive Maintenance
- Anomally Detection
- Asset Management

Example: Circuit Breaker Service Life Estimation



Data Collection/Traning



Appropriate Algorithm/
Model Optimization



Remaining Operations:
1325

Regression Result

Smart Portable Test Unit

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The SPTU can provide value-added services, such as pre-delivery testing validation and routine operational testing while helping to accelerate the market scale of ACB and MCCB products.

Feature

Compatible with all CHINT ACB, including NA1 NXA and NA8 series.

Value

Strong compatibility, reduce customer costs.

Benefit

Only one test device is needed to complete all the CHINT brand ACB tests.

Feature

Online detection continuously

Advantage

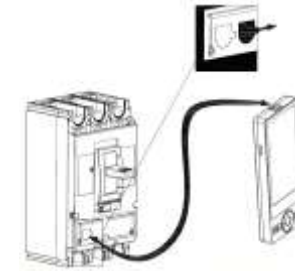
on-site testing rapidly and Conveniently

Benefit

Ensure power supply continuously



Appearance image



MCCB
Connection by ETH



ACB
Connection by USB

Back

SPTU with advanced field maintenance function

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Support data collection and parameter setting functions for grid connected circuit breakers, achieve local operation and maintenance, health detection, protection replacement, online monitoring, and remote upgrading functions, and perform "one key sequential control" function on switches to assist in the digital transformation and lean operation and maintenance of distribution equipment.

Functions

HMI

Support touchable screen HMI and customize APP function applications

Data security

Connect securely to the internal network through USB encryption, upload on-site data to the servers for analysis and processing, or download data from the internal network.

Protection substitution

Protection functions can be executed continuously when replacing plug-in modules of breaker, thanks for SPTU's taking over of fault analysis and action

Health self check

Comprehensive self inspection of the health status of grid connected switches by excitation fault signal to it

wireless communication

Supports 4G communication and wireless public/private network

Supports Bluetooth

Can quickly configure breaker through BLE

Diverse communication methods

Support multiple types of device access based on RS485 and CAN-FD

Support extension

Supports USB extension of DI, DO, AI, AO interfaces;

Easy to operate

Support front and back cameras to scan codes, take photos, and record on-site equipment operation information

Data storage

Support SD card storage of fault information

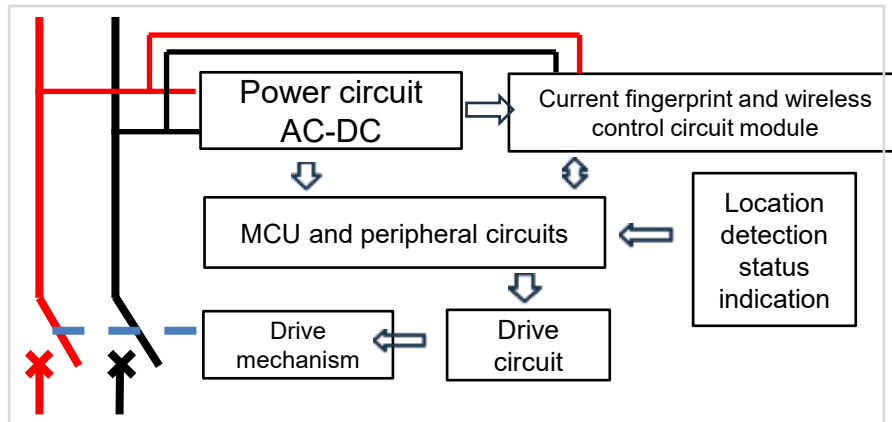


[Back](#)

Current Fingerprint Technology

Background : Inside metering box with wireless fee control method, the energy meter needs to match the circuit breaker for end user. The traditional method is to scan code for pairing. Usually the implementation personnel needs to find and pair each metering and circuit breaker one by one, which is heavy workload and easy to be wrong pair, and wrong pair can not even be detected in time, finally it may lead to electrical failure, like as cut off A's electricity but for B's bills was not paid.

Principle: Based on current spectrum, smart energy meter can identify the pair relationship between circuit breaker and smart energy meter by encrypted identity information of circuit breaker. After smart energy meter sends couple reactive signal with limited amplitude on the power line, it would get special current spectrum from circuit breaker, which is unique, then smart energy meter identify the circuit breaker, and the relationship can be guaranteed in real time.



Schematic block diagram



Fingerprint current waveform

Value:

- Power line transmission signal, no complicated wiring
- No human intervention required, automatic networking, reducing labor costs and error rates

Expand

- [Control command execution authorization](#), use current fingerprint to authorize control commands in scenarios that require control .
- Draw lessons from the principle of ultrasonic flaw detection, inject current fingerprint information into the line, and collect the current fingerprint information after flowing through the line for comparison and analysis [to monitor line faults and determine the fault location](#).

Definition: Islanding is a condition that occurs when a distributed energy resource (DER) such as a grid-tied inverter continues to supply power to a section of the grid that has been disconnected from the main grid.

Importance of Islanding Detection

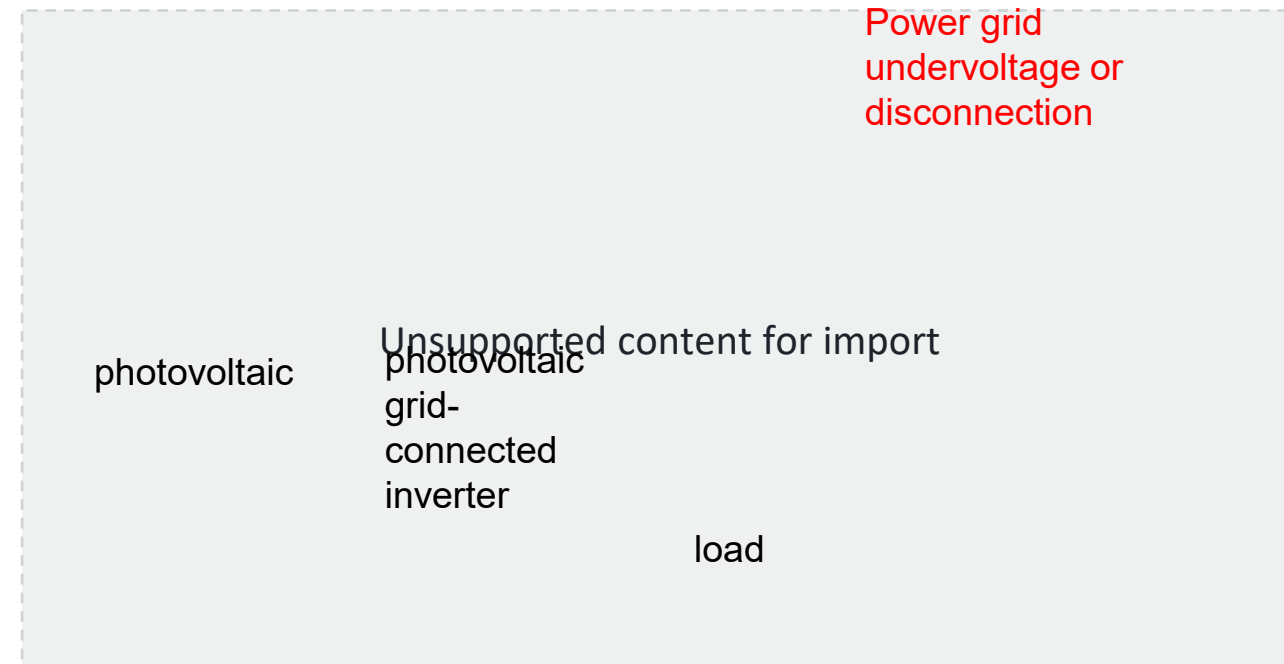
Safety: Islanding can pose significant safety hazards for utility workers and the public. If a disconnected section of the grid continues to receive power from a DER, it can lead to electrical shock, electrocution, and even fire hazards.

Equipment damage: Islanding can also cause damage to the equipment in the grid-tied inverter and the electrical grid. Continued power supply to the disconnected section of the grid can cause equipment failure, leading to costly repairs and downtime.

Grid stability: Islanding can lead to a complete collapse of the electrical grid, causing widespread power failures and disrupting essential services. Islanding detection ensures that the DER system is shut down in a timely manner to prevent any instability in the electrical grid.

Compliance: Many countries and regions have regulations and standards in place that require islanding detection in DER systems, such as IEEE 1547-2018. Compliance with these regulations is essential to ensure that the DER system is safe and reliable.

Reliability: Islanding detection methods ensure that the DER system is operating correctly and provides reliable power supply to the electrical grid. Timely detection of islanding and shutdown of the DER



Islanding protection

Types of Islanding Detection

There are generally two types of islanding detection methods used in grid-tied inverters:

1. Passive methods: These methods use the inherent characteristics of the electrical grid to detect islanding, **such as under/over voltage, frequency shift, rate of change of frequency and voltage phase jump etc.** Passive methods do not require any additional hardware or communication between the inverter and the electrical grid.

2. Active methods: Active islanding detection techniques involve the generation of disturbances in the electrical grid that the inverter system is connected to, and then analyzing the output voltage or frequency of the inverter to investigate to check whether there is islanding. Active methods can be more accurate and reliable than passive methods, but they require additional costs and complexity.

